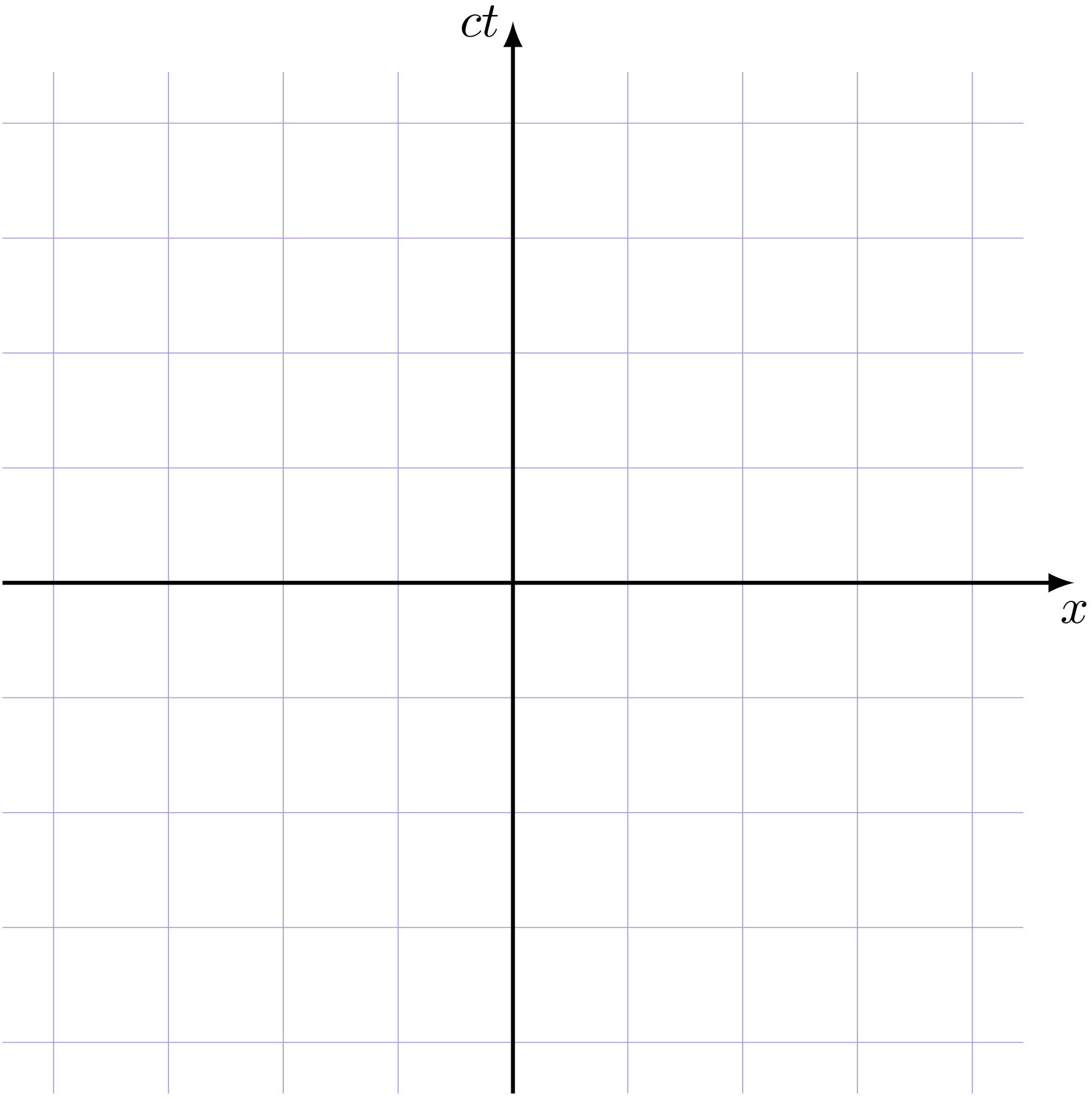
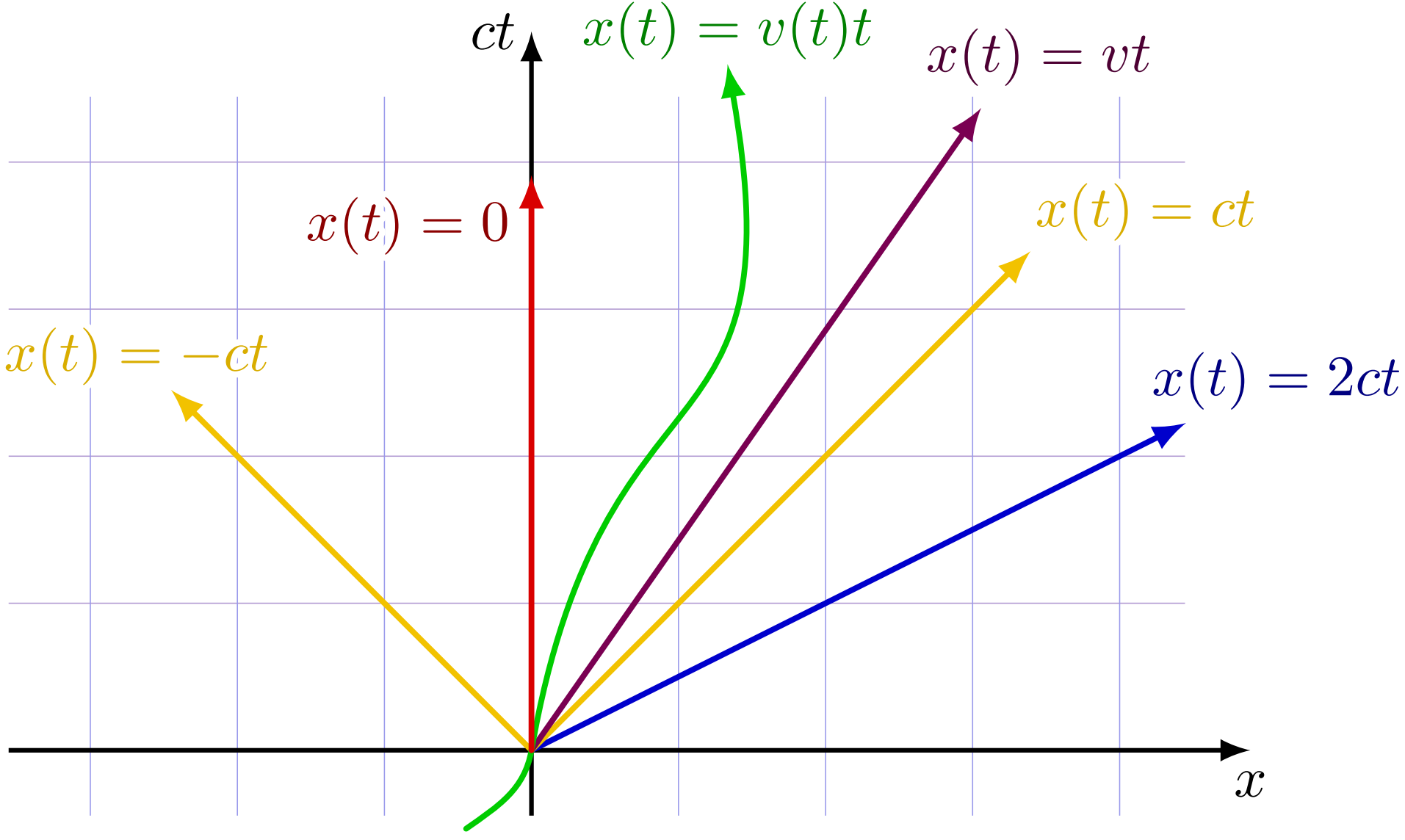
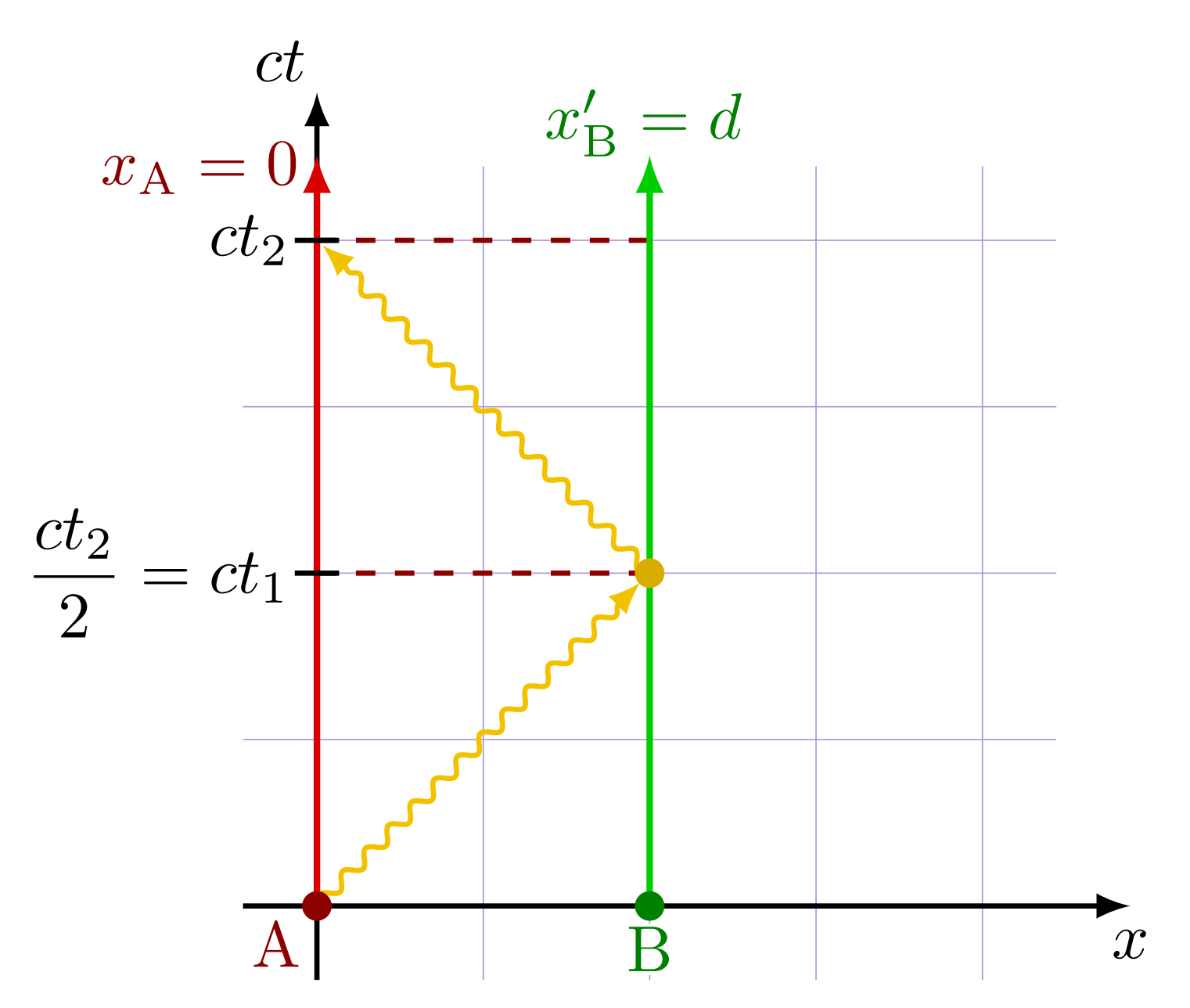
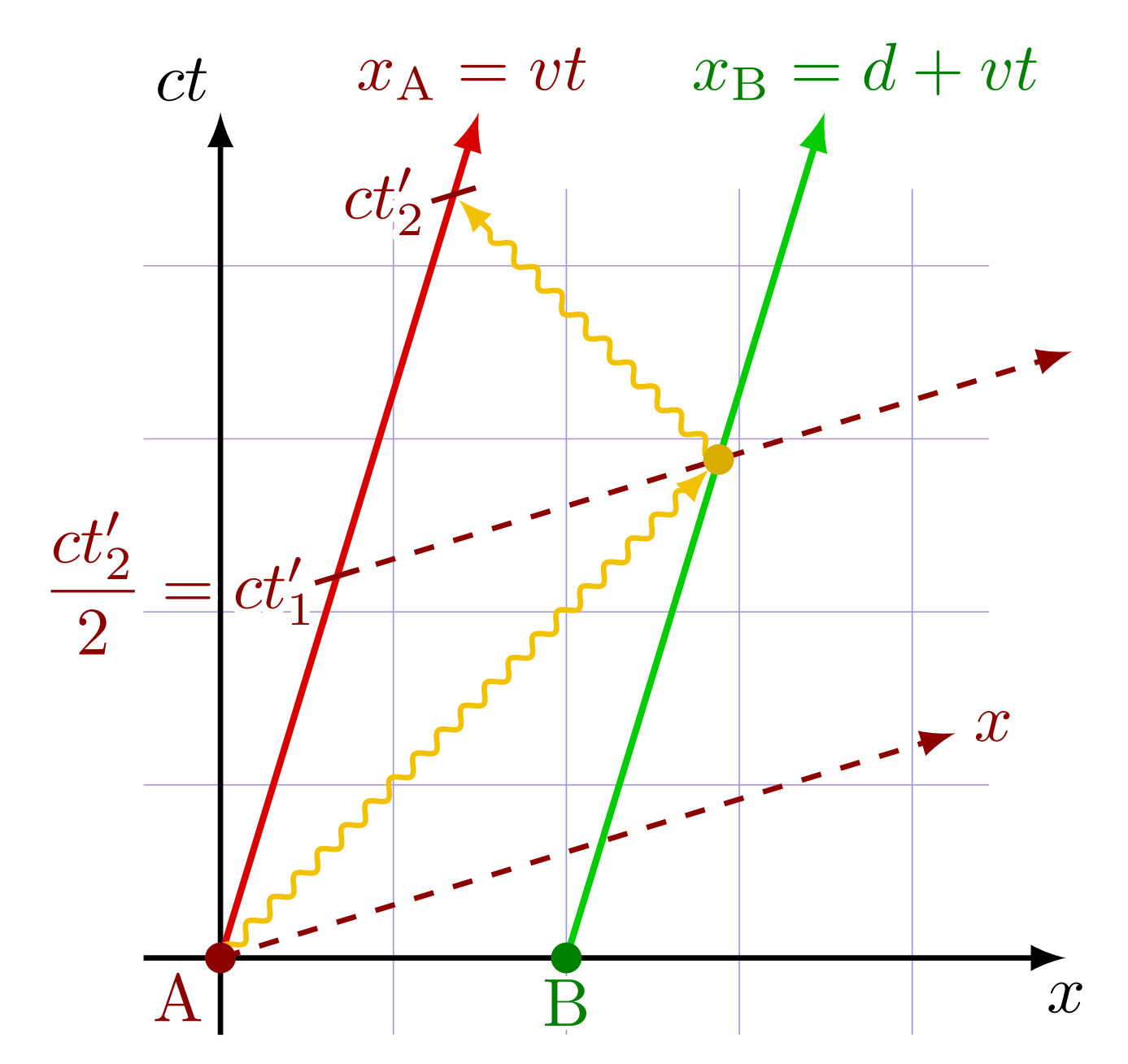
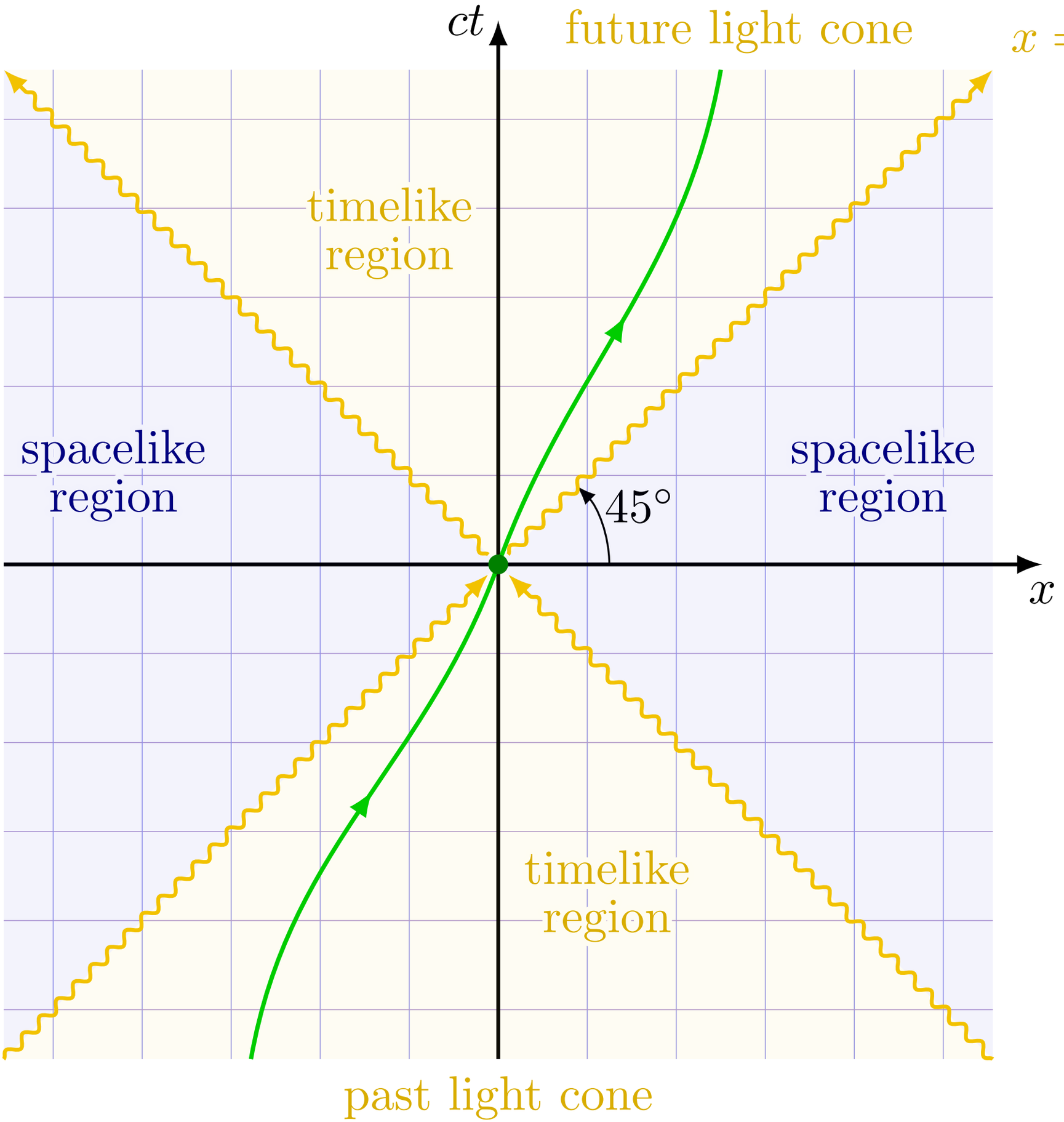
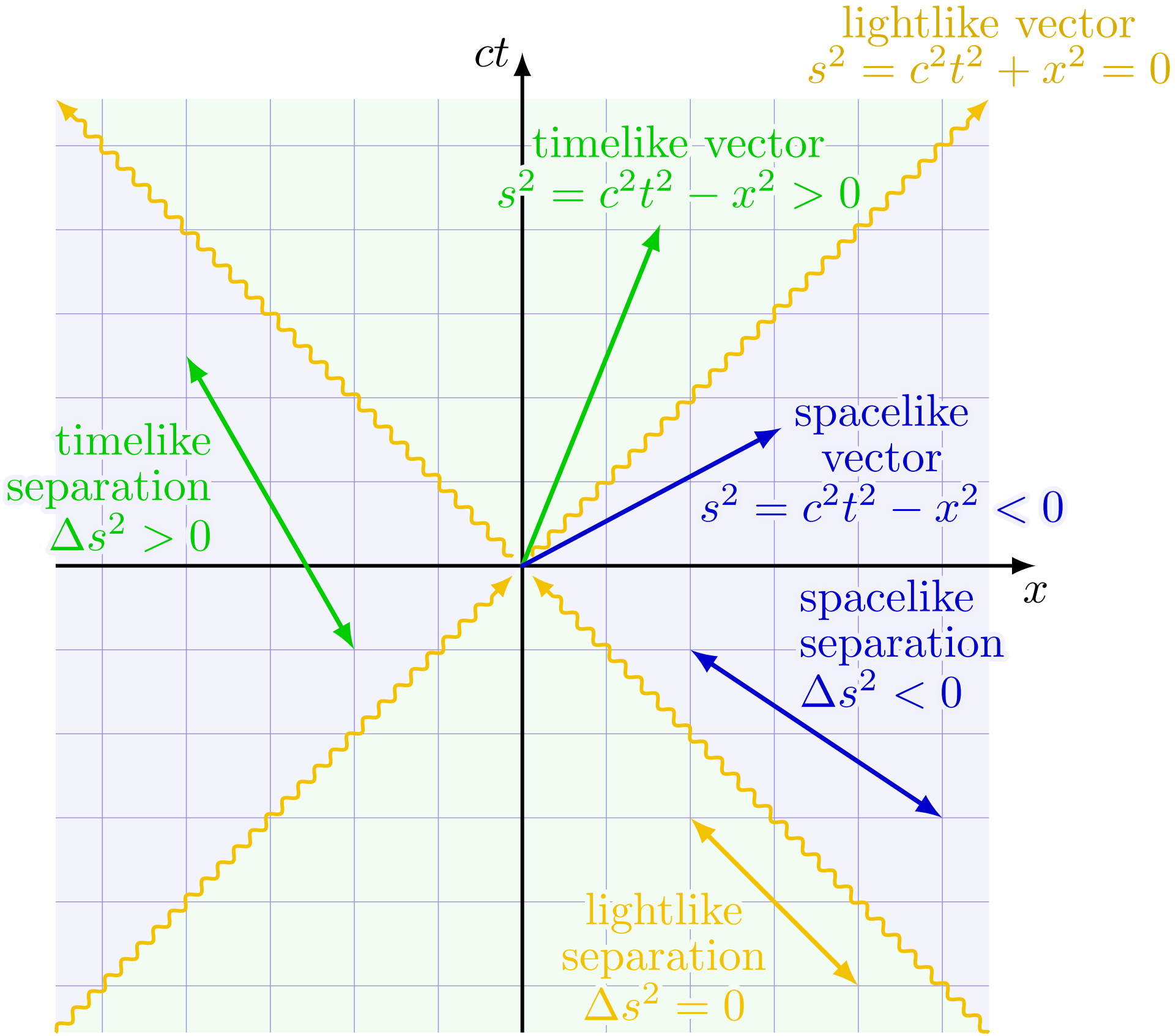

ct
x

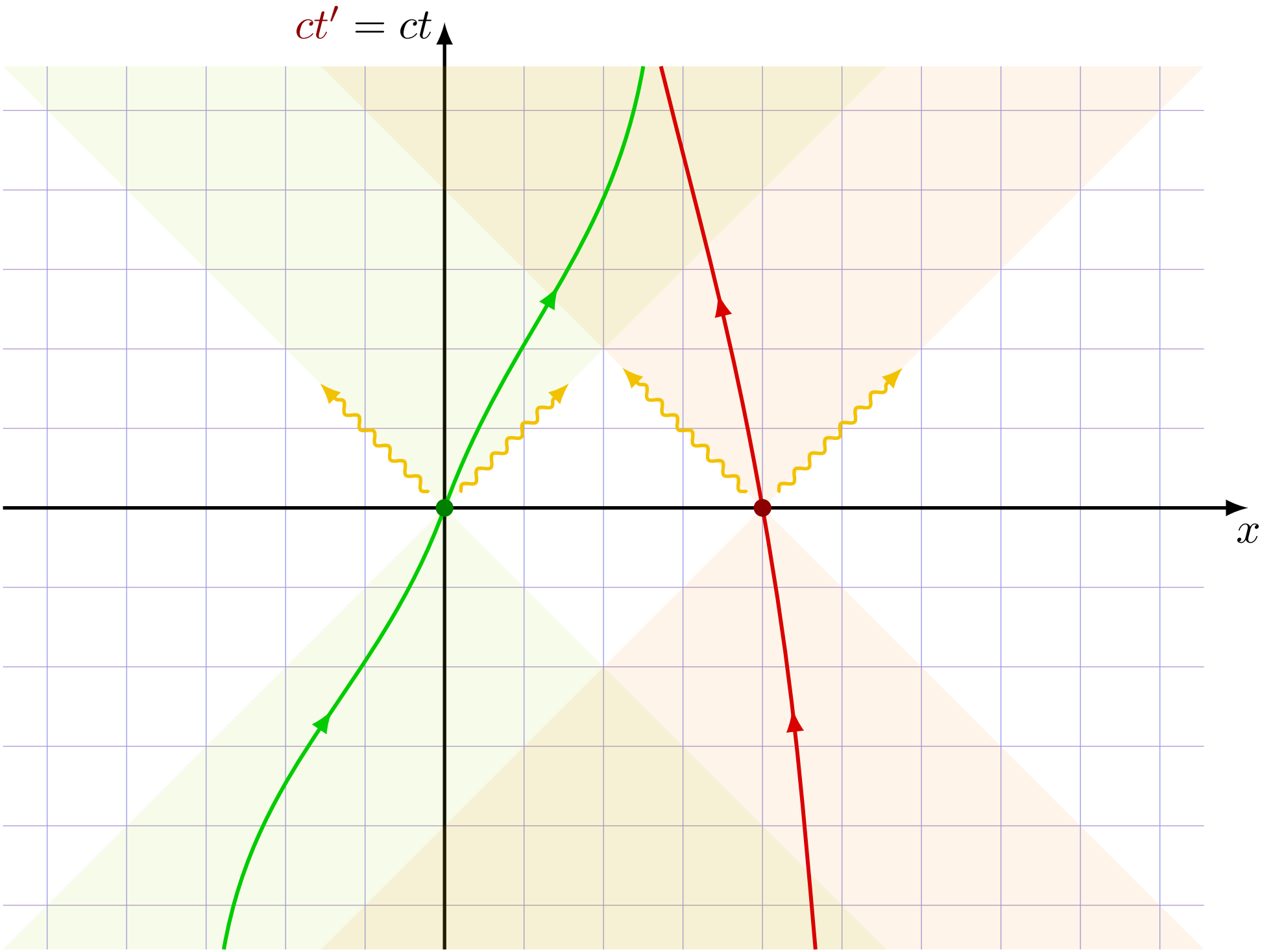

ct
$x(t) = v(t)t$
$x(t) = vt$
$x(t) = 0$
$x(t) = ct$
$x(t) = -ct$
$x(t) = 2ct$
x

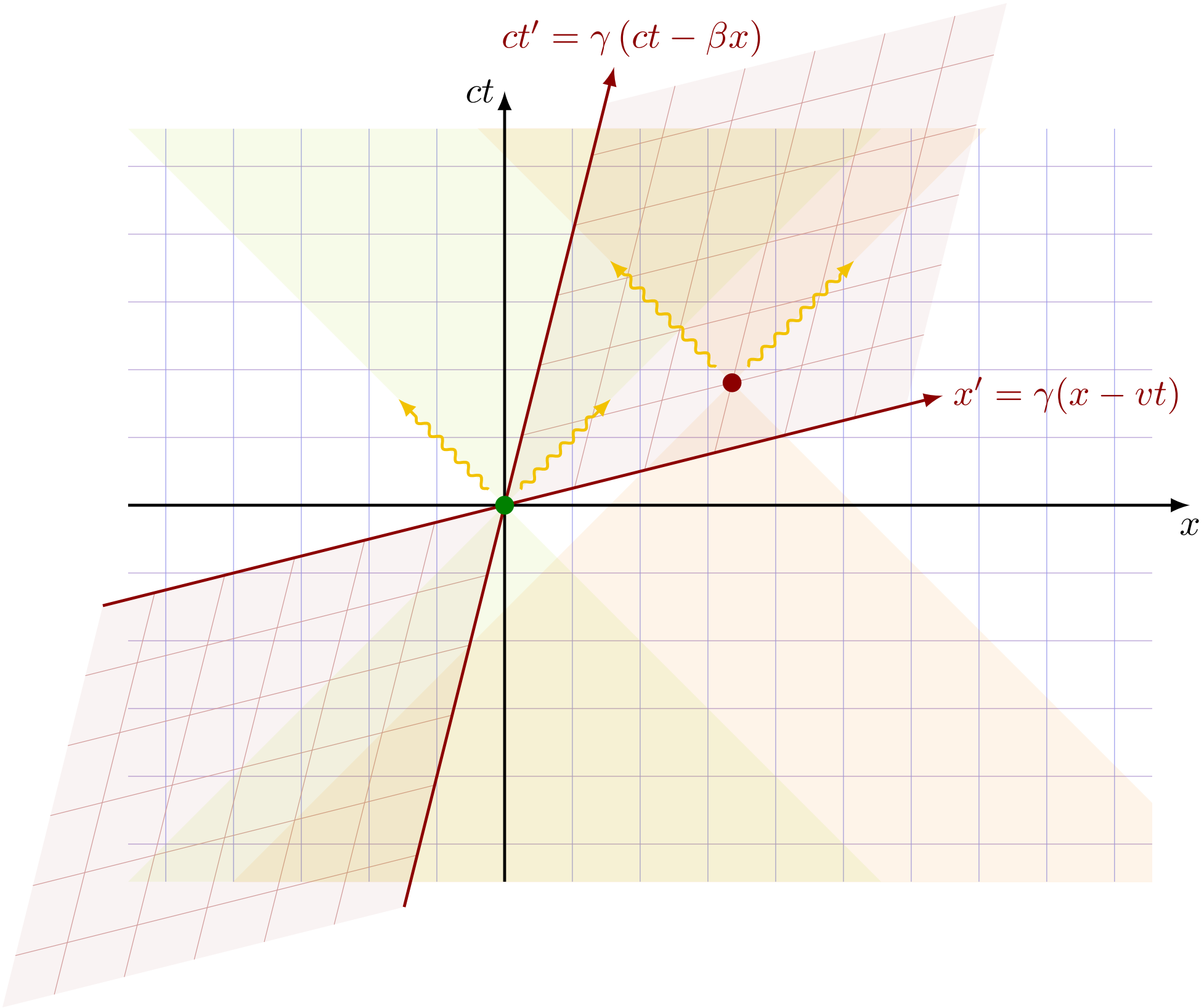

ct
$x_\mathrm{A} = 0$
$x'_\mathrm{B} = d$
ct_2
$\dfrac{ct_2}{2} = ct_1$
A
B
x

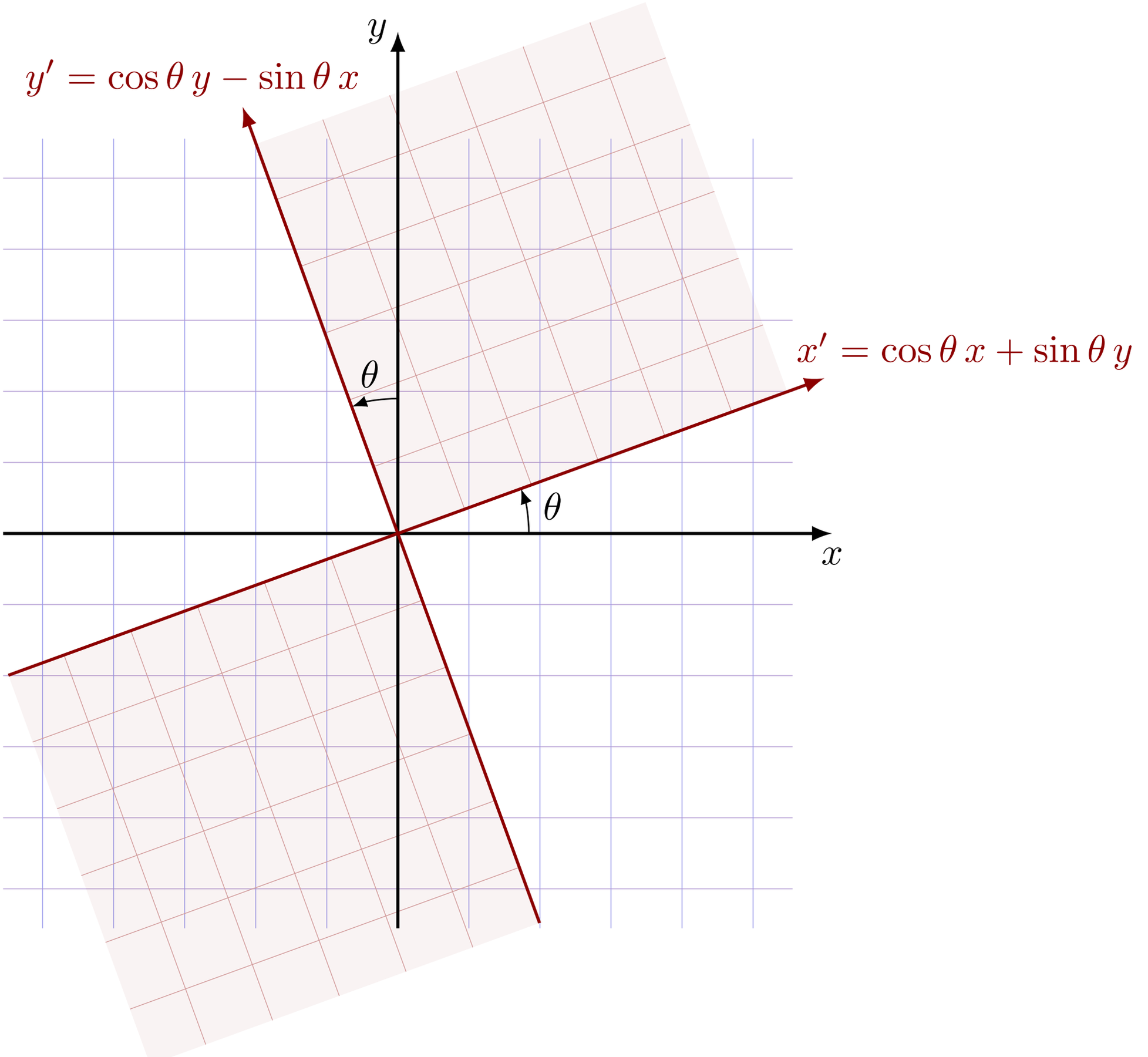

ct
$x_\mathrm{A} = vt$
$x_\mathrm{B} = d + vt$
ct'_2
$\frac{ct'_2}{2} = ct'_1$
x
A
B
x

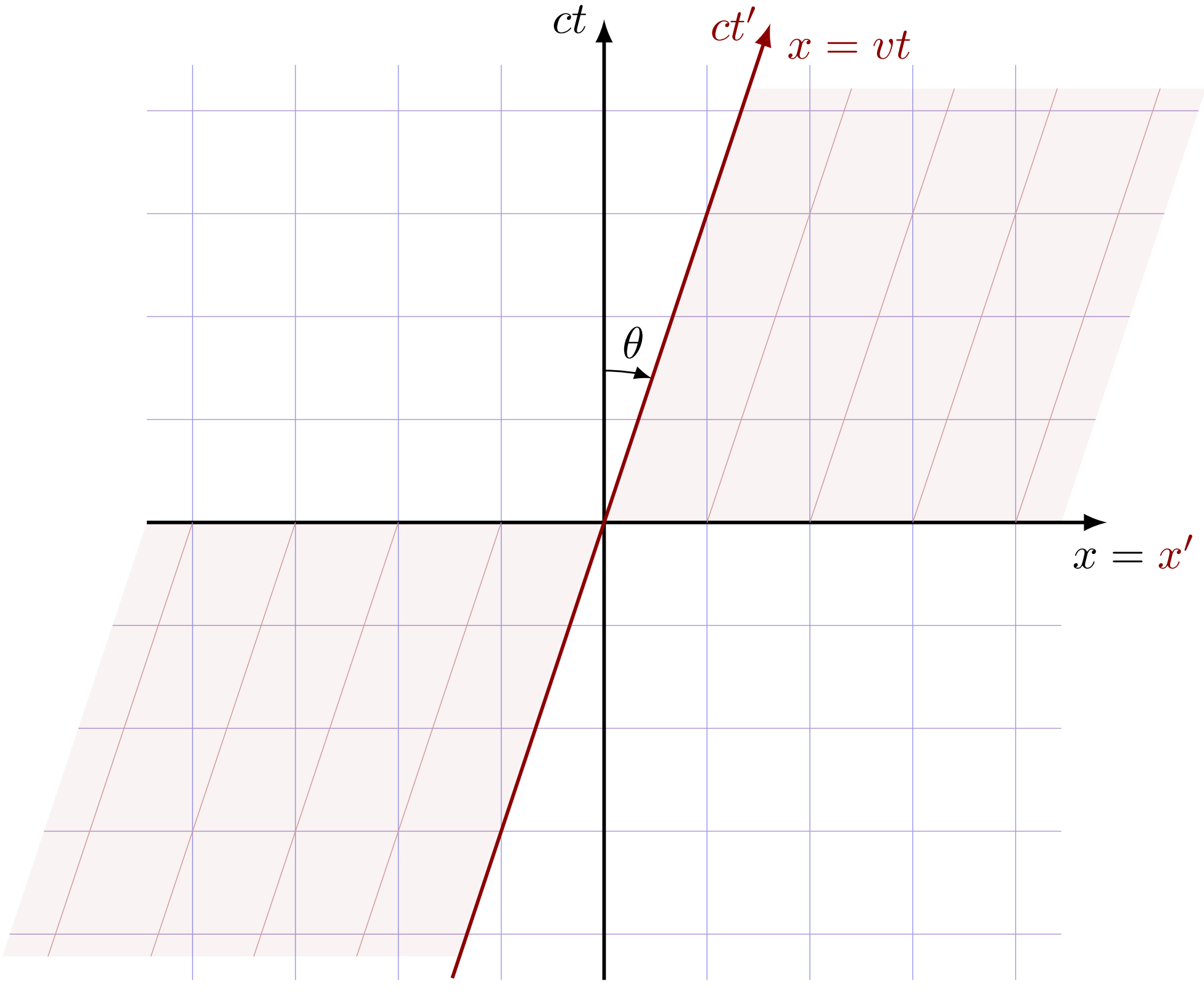

ct
future light cone
$x = ct$
timelike region
spacelike region
spacelike region
45°
x
timelike region
past light cone

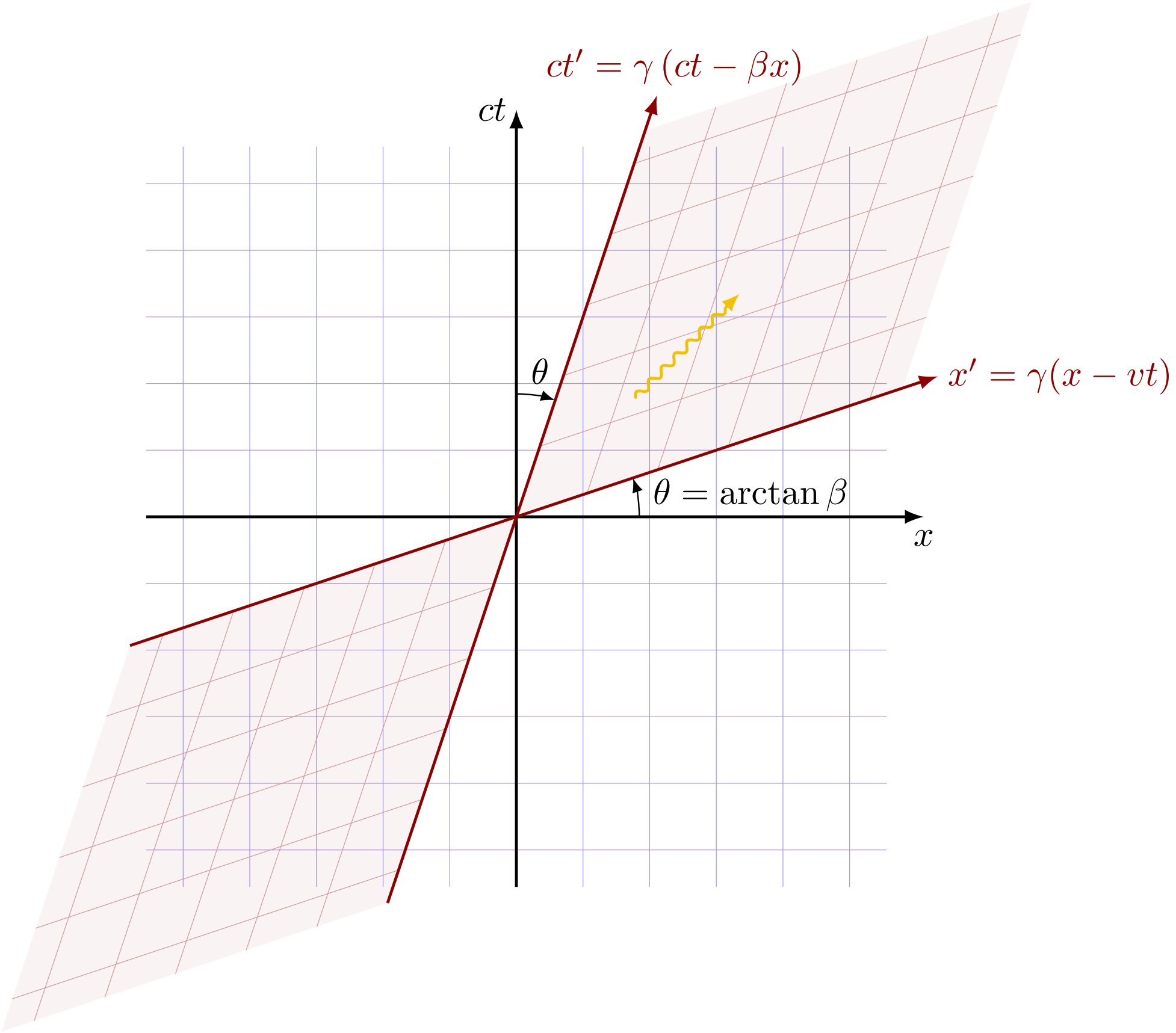

lightlike vector
$s^2 = c^2t^2 + x^2 = 0$
ct
timelike vector
$s^2 = c^2t^2 - x^2 > 0$
timelike
separation
$\Delta s^2 > 0$
spacelike
vector
$s^2 = c^2t^2 - x^2 < 0$
x
spacelike
separation
$\Delta s^2 < 0$
lightlike
separation
$\Delta s^2 = 0$

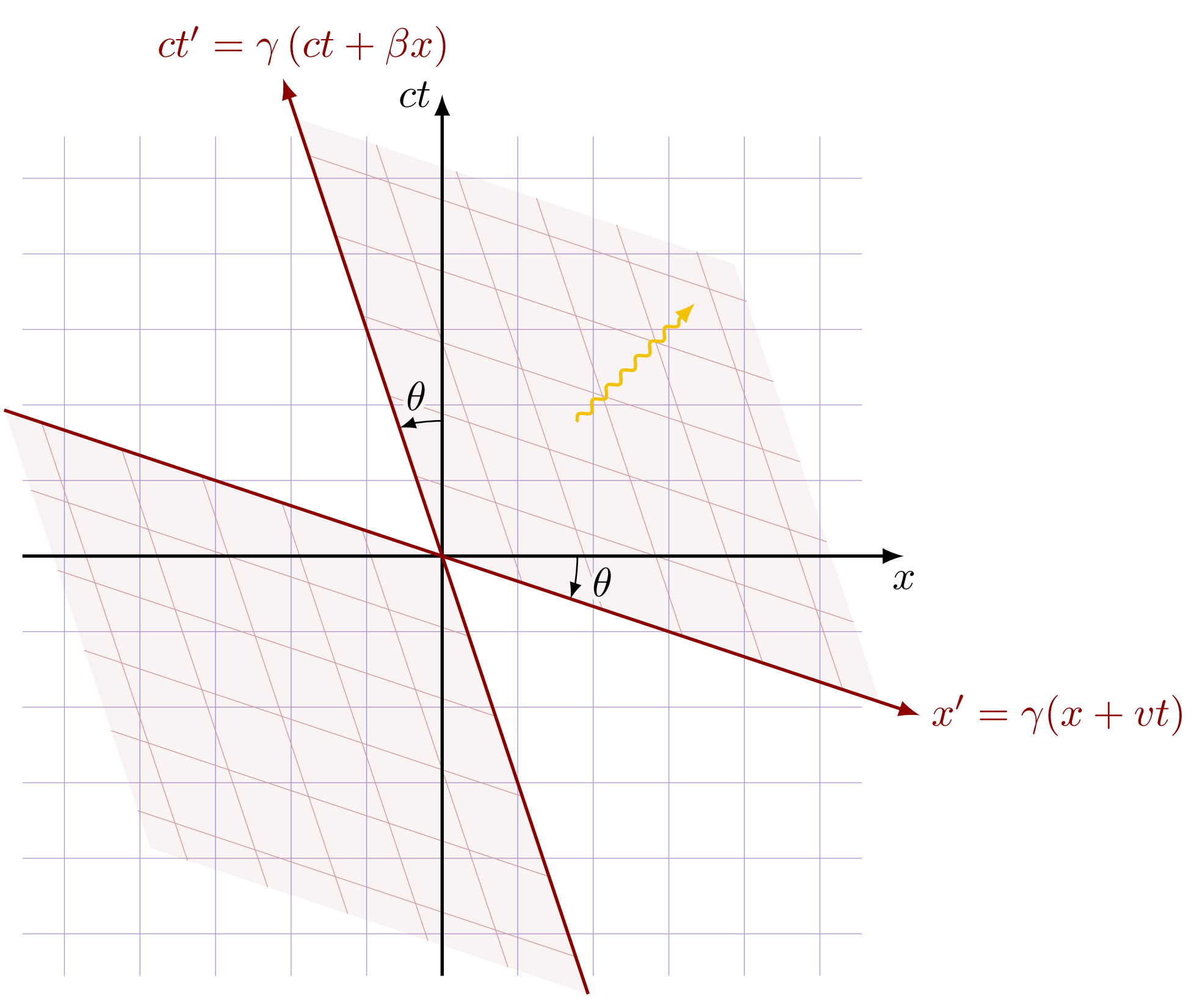

$ct' = ct$
x

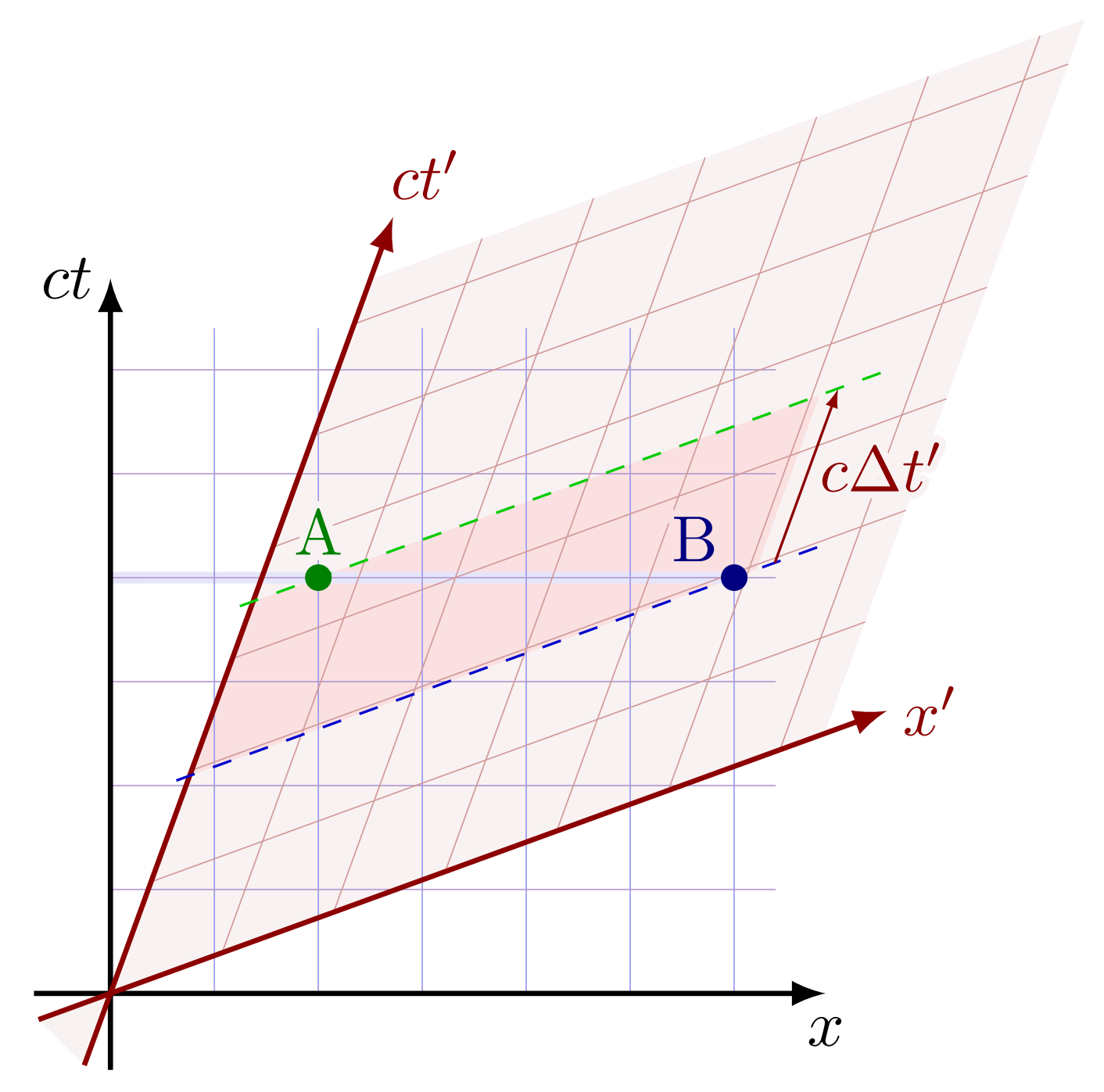

$ct' = \gamma\,(ct - \beta x)$
ct
$x' = \gamma(x - vt)$
x

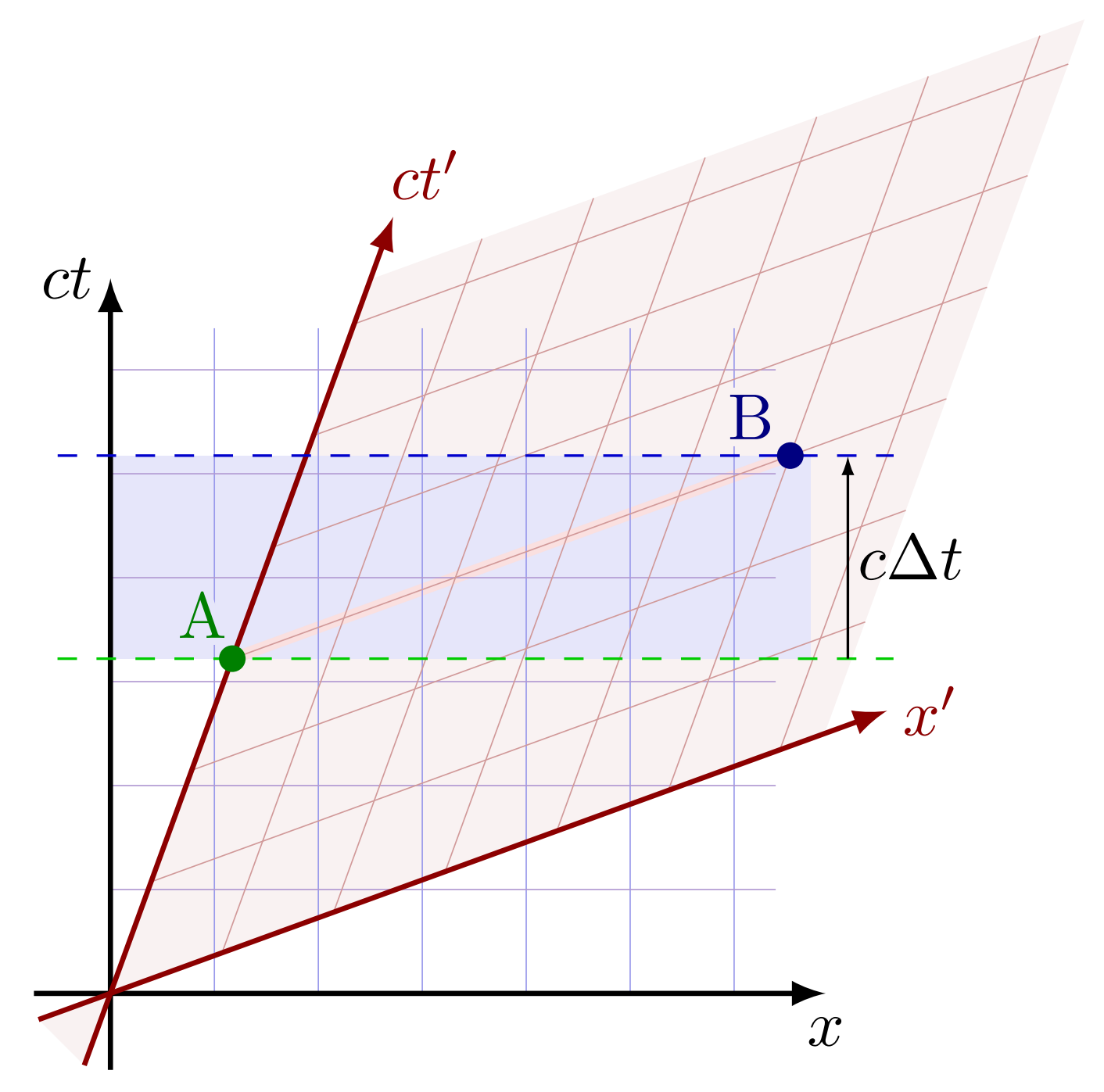

y
$y' = \cos\theta\, y - \sin\theta\, x$
$x' = \cos\theta\, x + \sin\theta\, y$
θ
θ
x

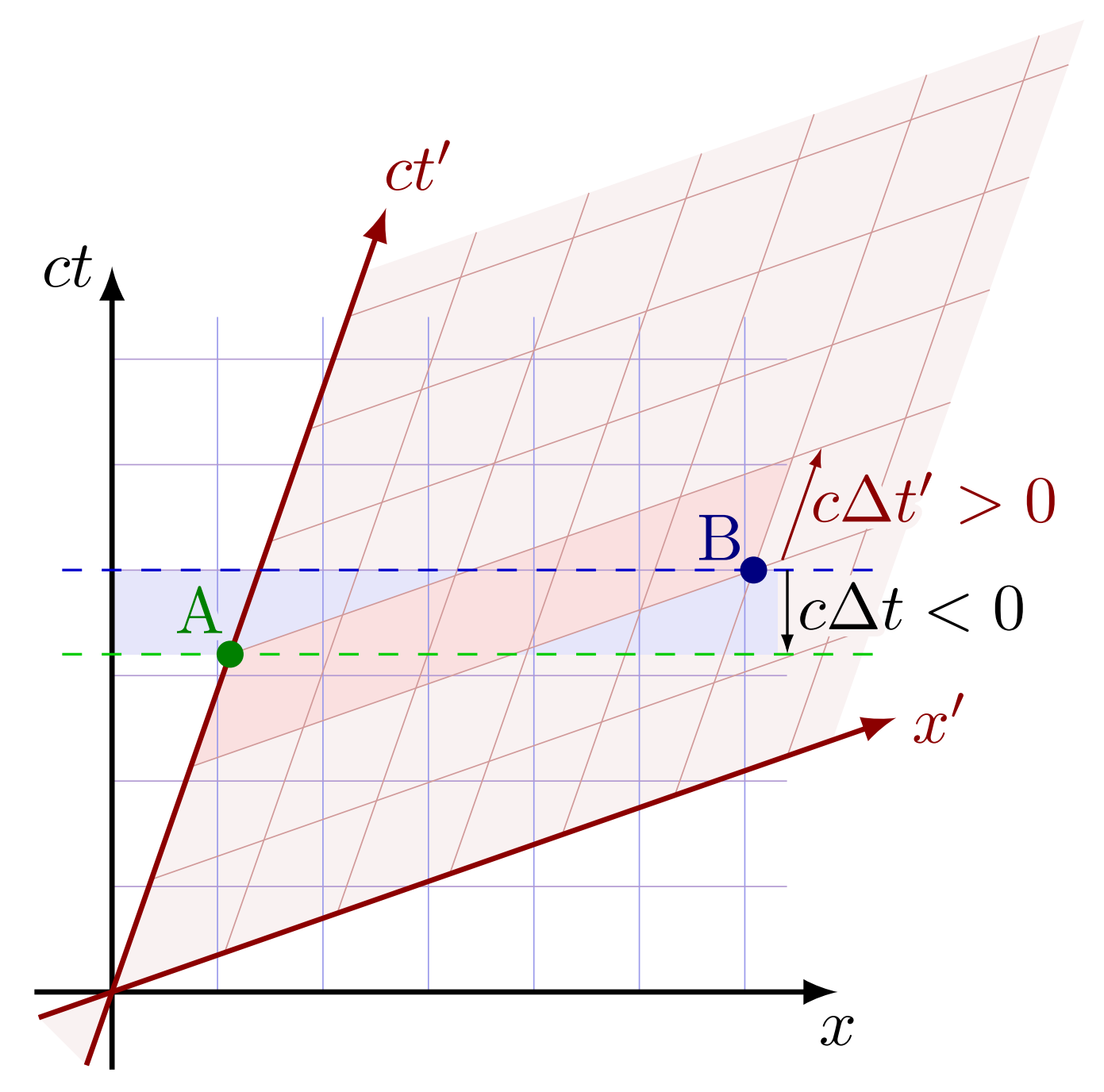

ct
ct'
$x = vt$
θ
$x = x'$

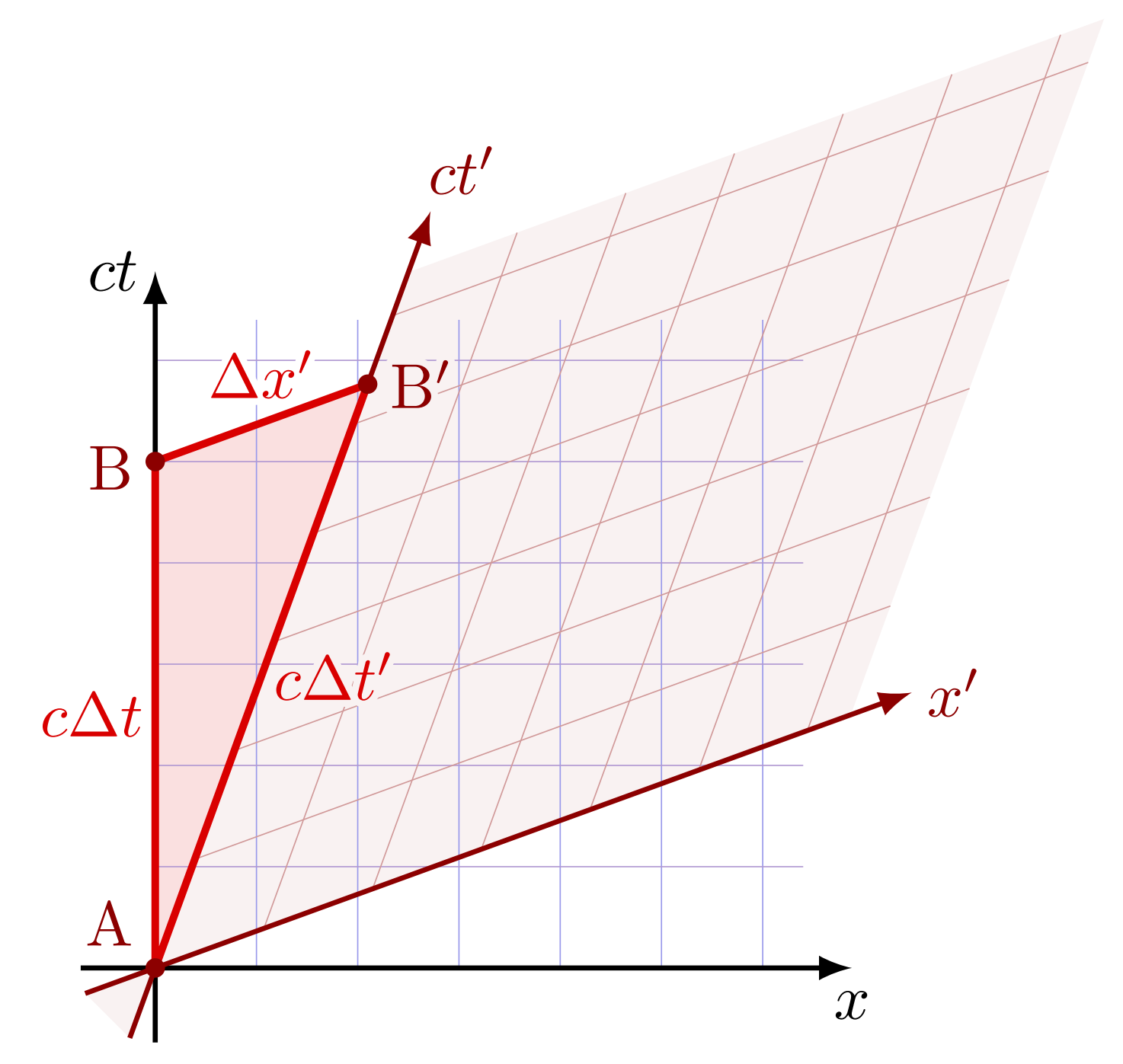

$ct' = \gamma\,(ct - \beta x)$
ct
θ
$x' = \gamma(x - vt)$
$\theta = \arctan\beta$
x

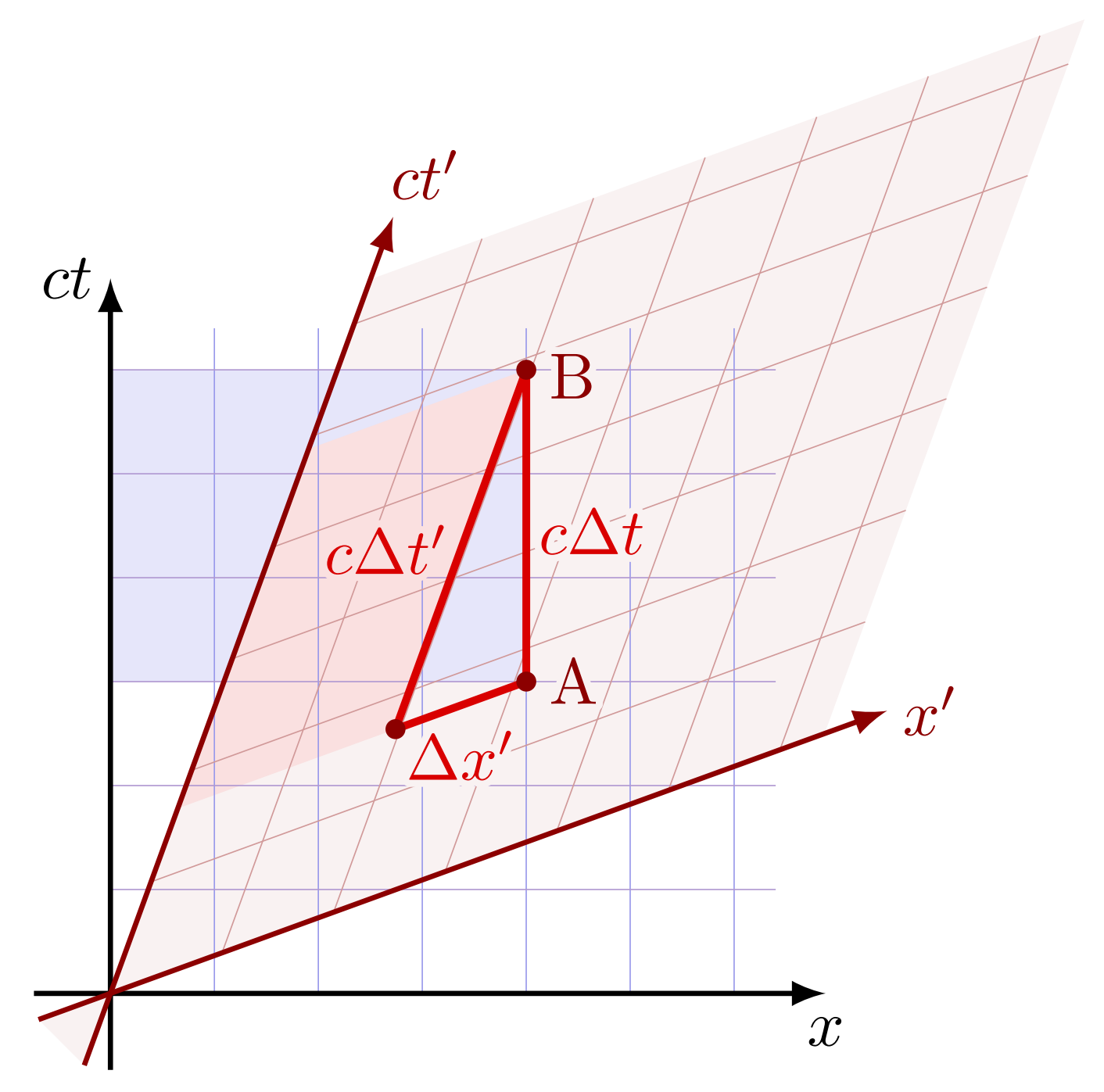

$ct' = \gamma\,(ct + \beta x)$
ct
θ
θ
x
$x' = \gamma(x + vt)$

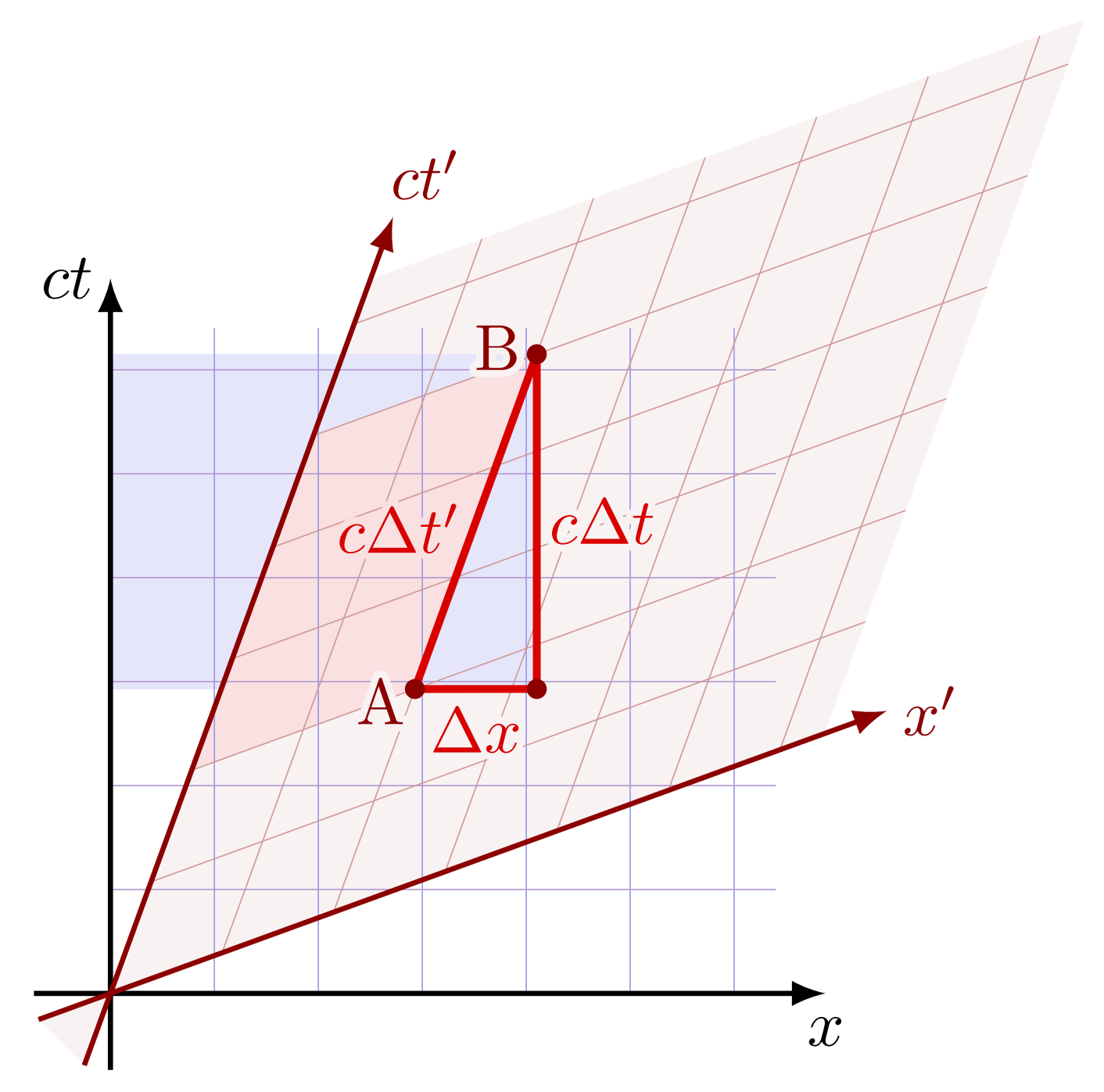

ct'
ct
A
B
$c\Delta t'$
x'
x

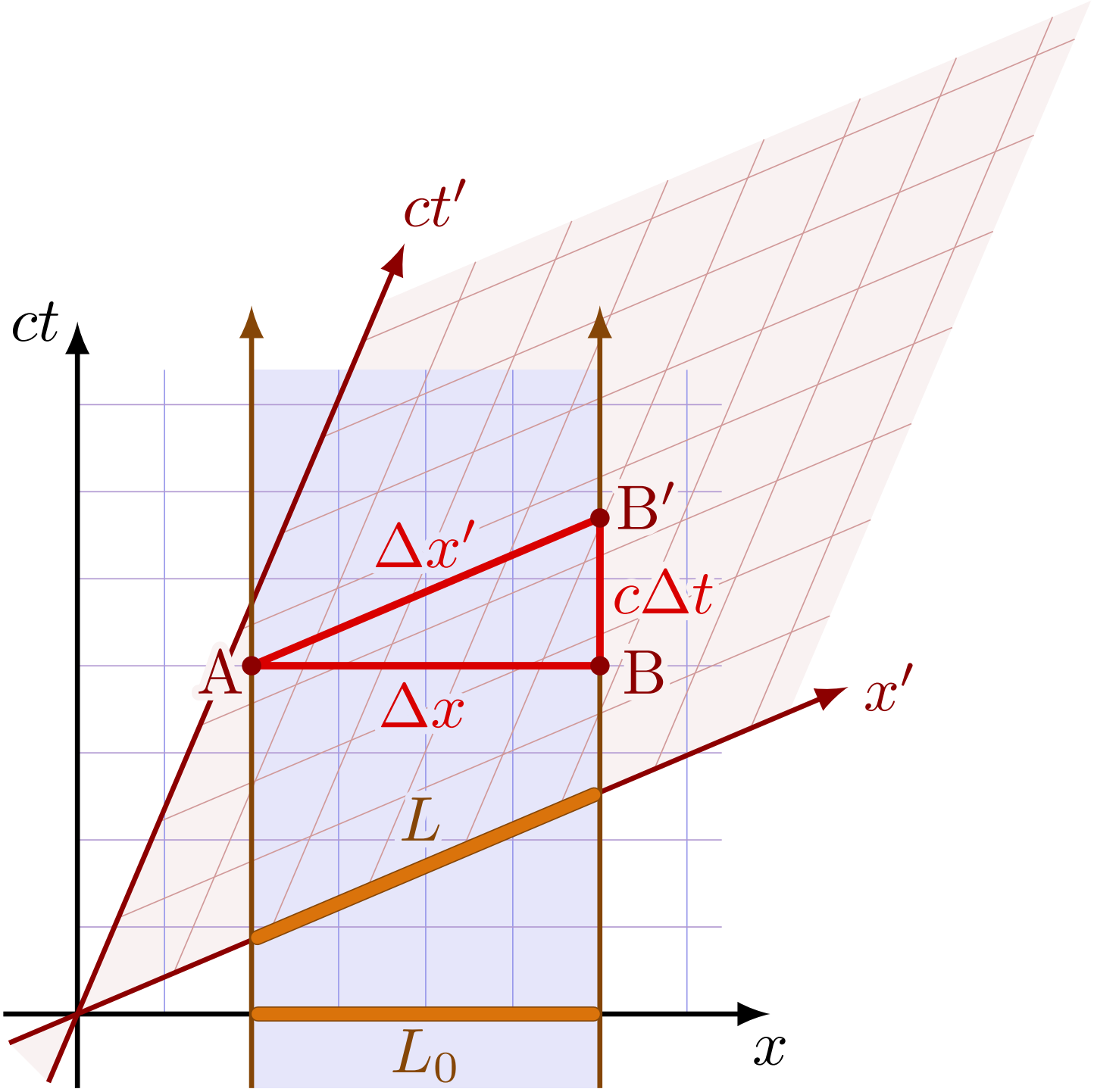

ct'
ct
B
A
$c\Delta t$
x'
x

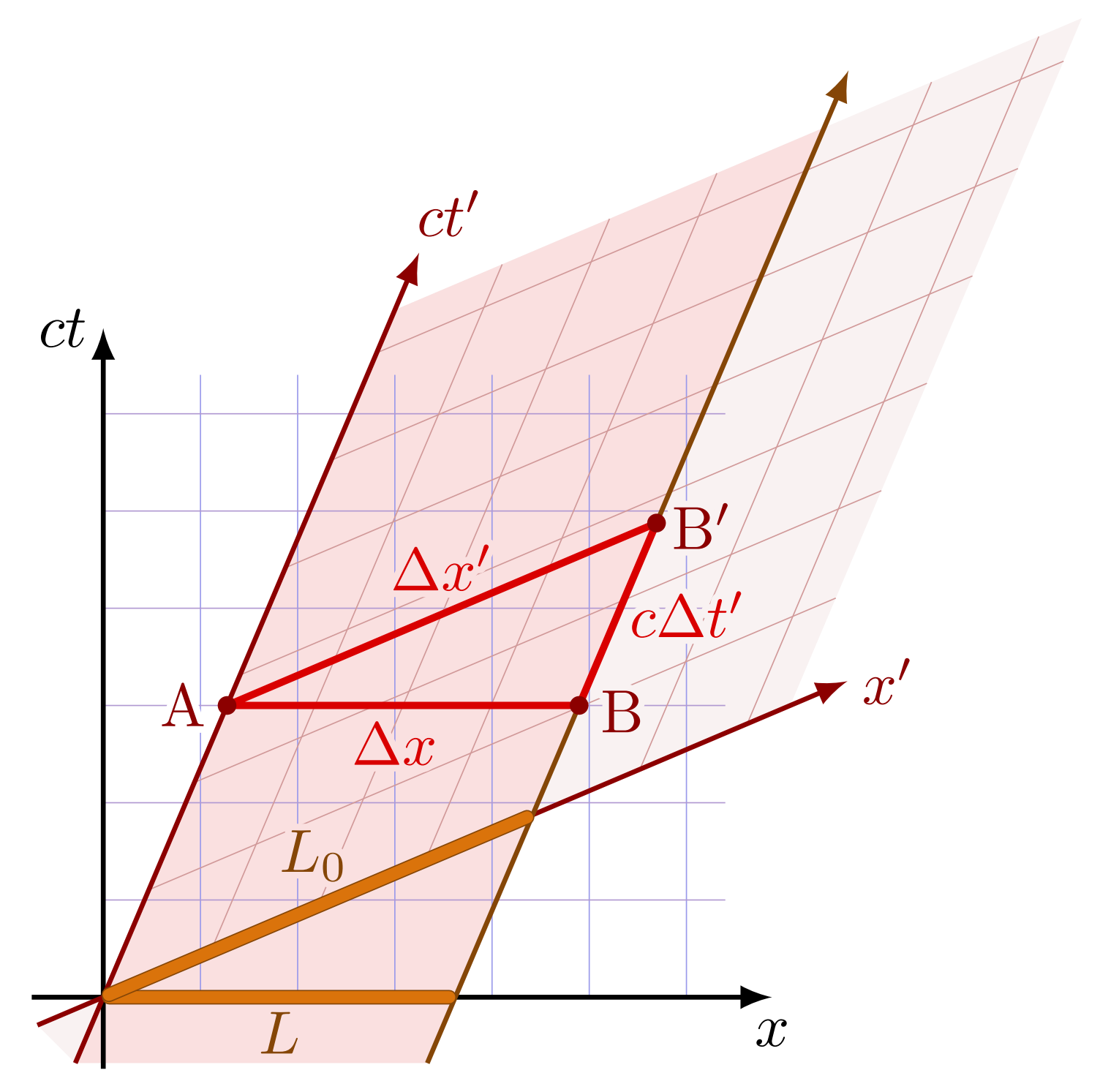

ct'
ct
$c\Delta t' > 0$
B
A
$c\Delta t < 0$
x'
x

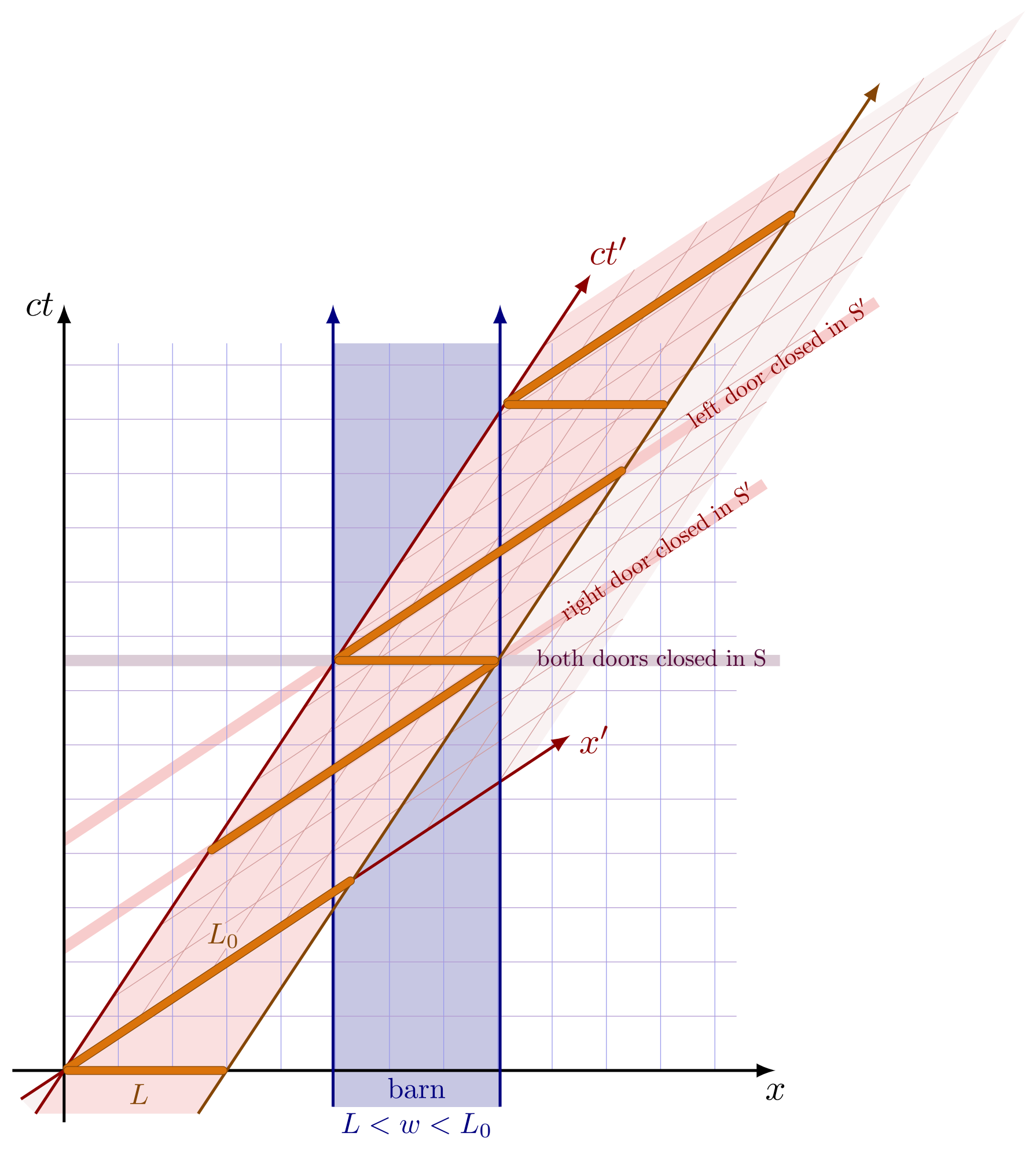

ct'
ct
$\Delta x'$
B′
B
$c\Delta t'$
$c\Delta t$
x'
A
x

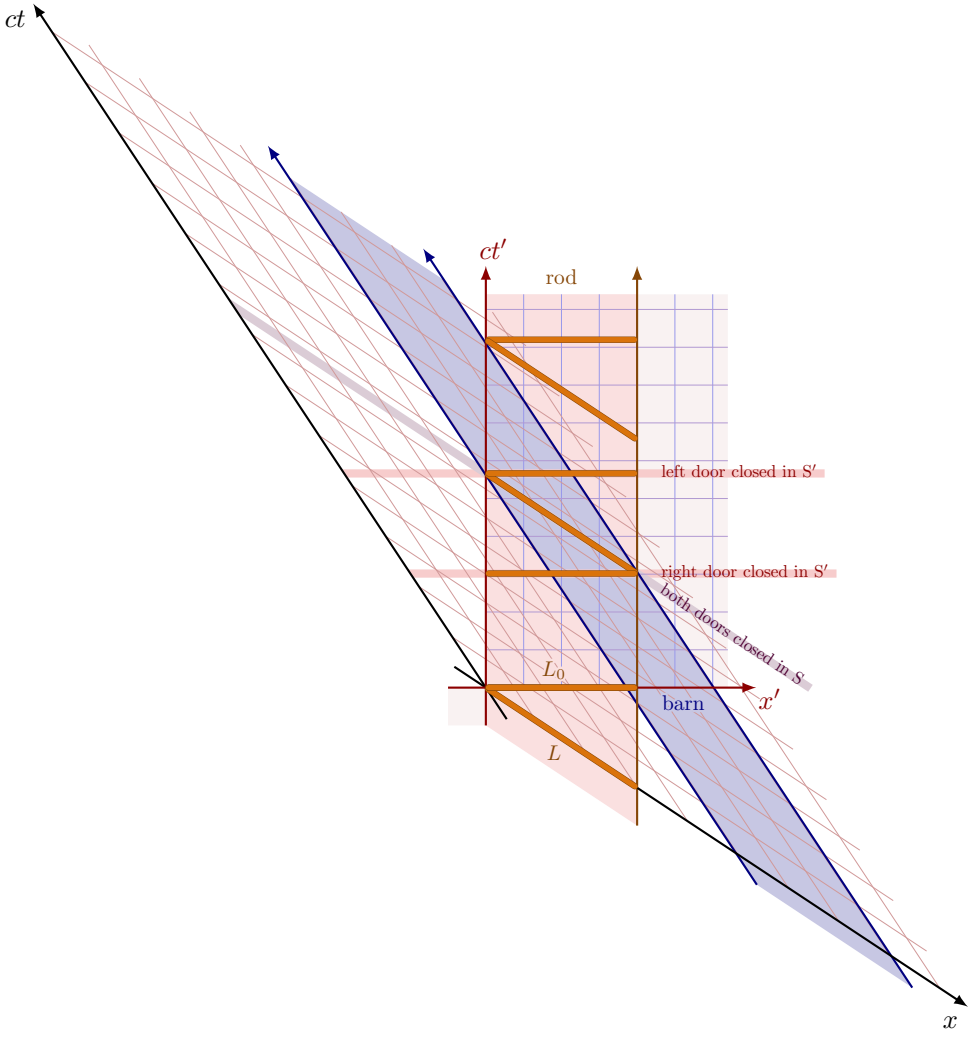

ct'
ct
B
$c\Delta t'$
$c\Delta t$
A
x'
$\Delta x'$
x

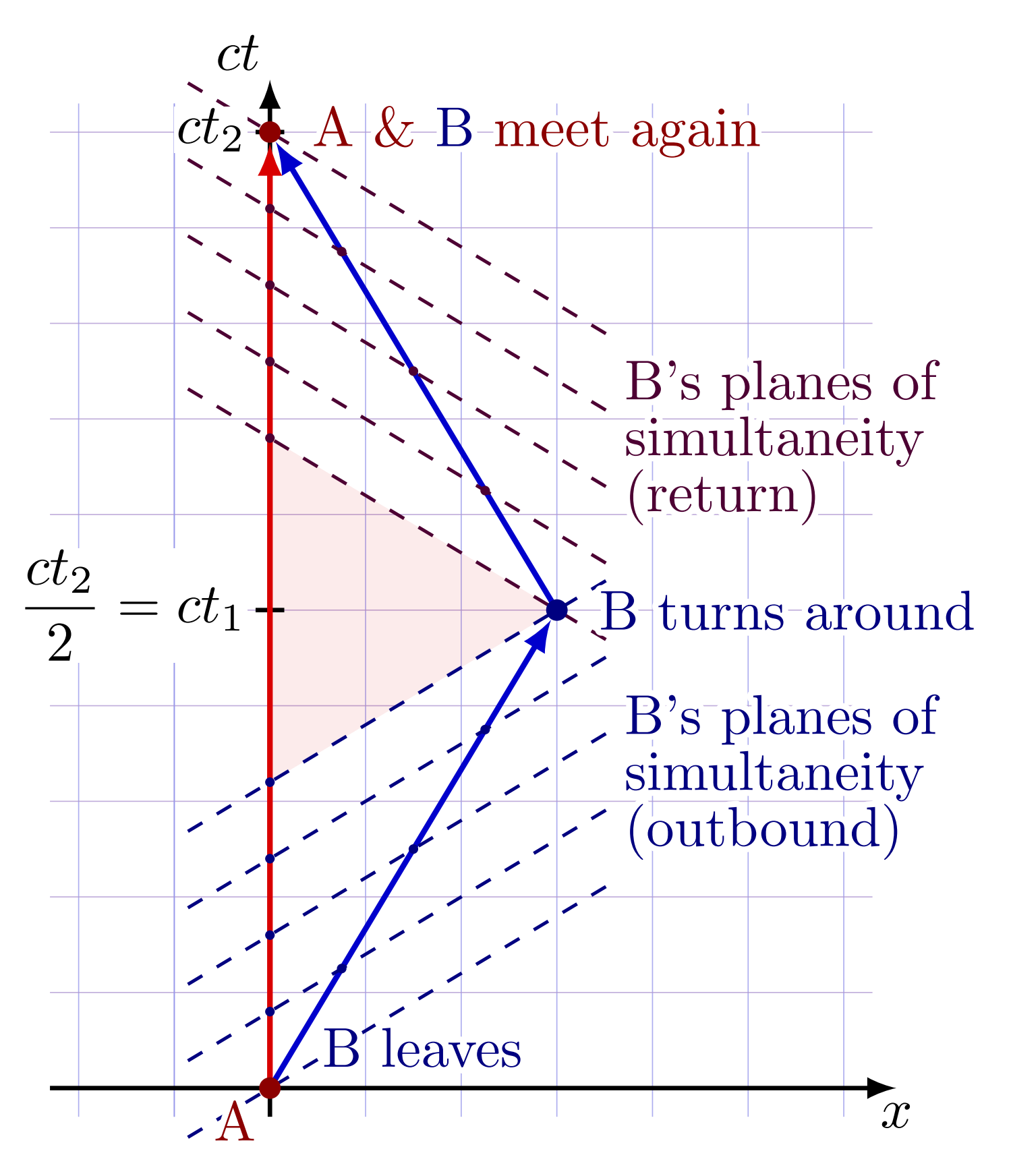

ct'
ct
B
$c\Delta t'$
$c\Delta t$
A
Δx
x'
x

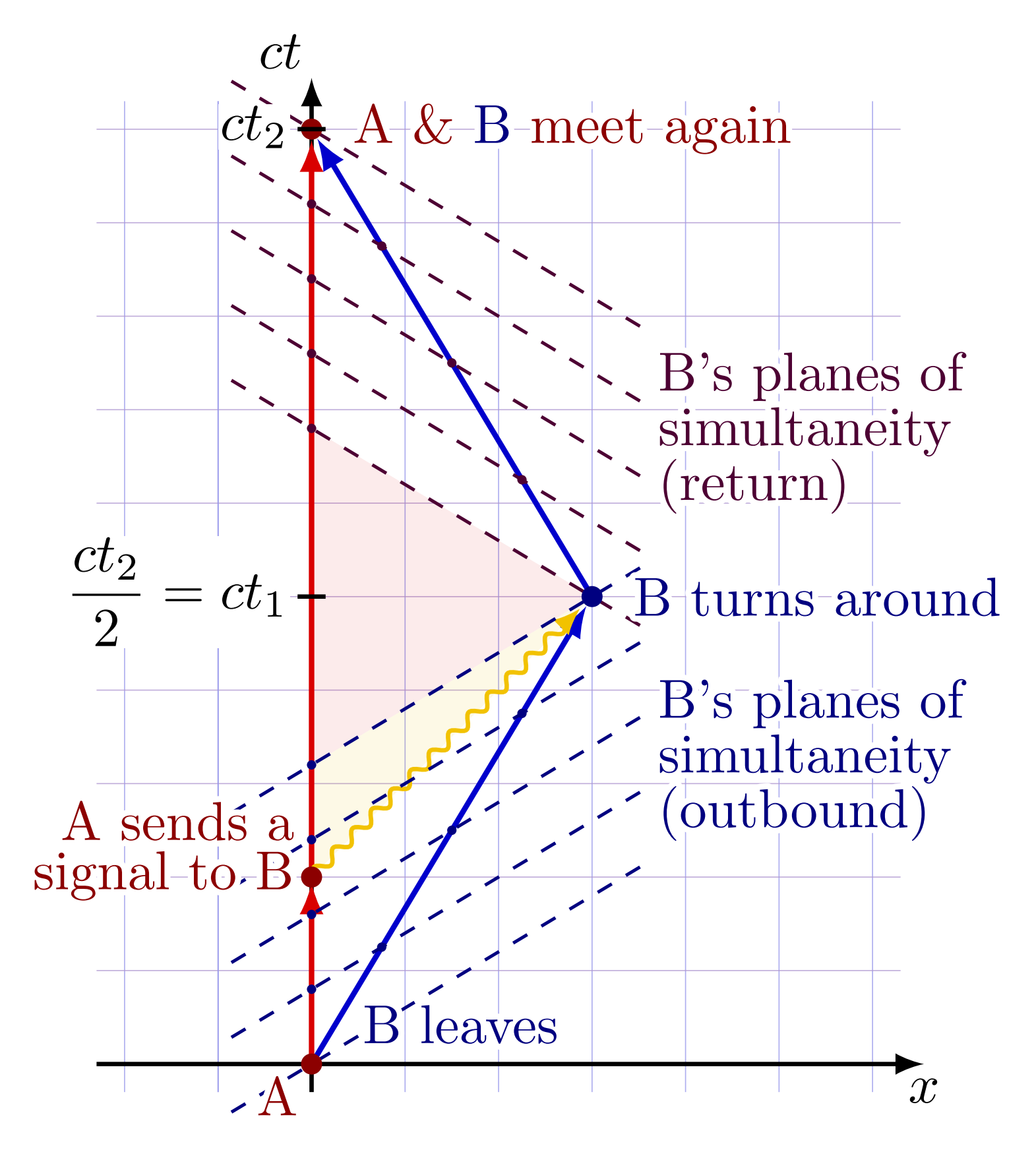

ct'
ct
B′
$\Delta x'$
$c\Delta t$
A
B
Δx
x'
L
L_0
x

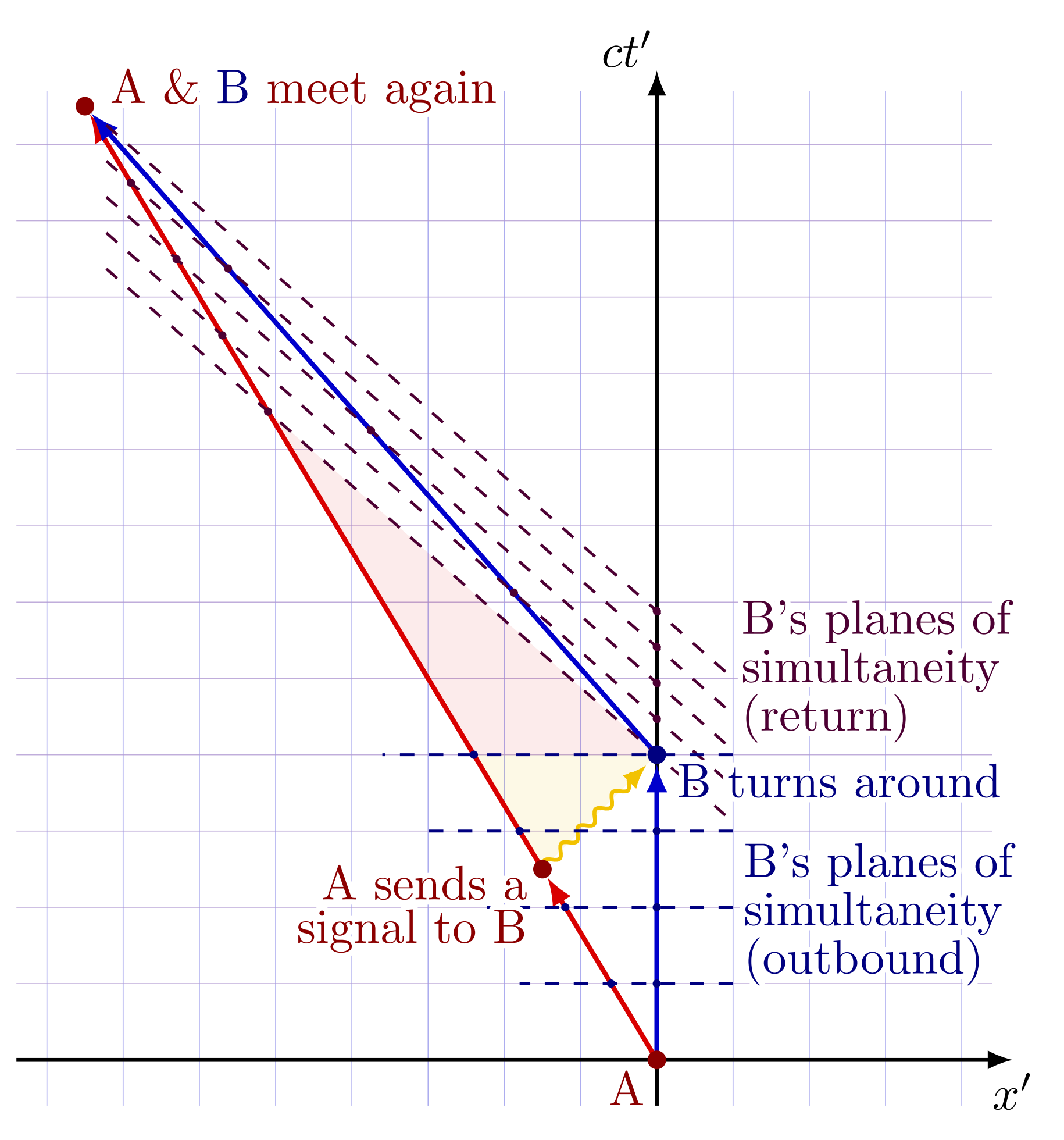

ct'
ct
B′
$\Delta x'$
$c\Delta t'$
A
B
x'
Δx
L_0
L
x

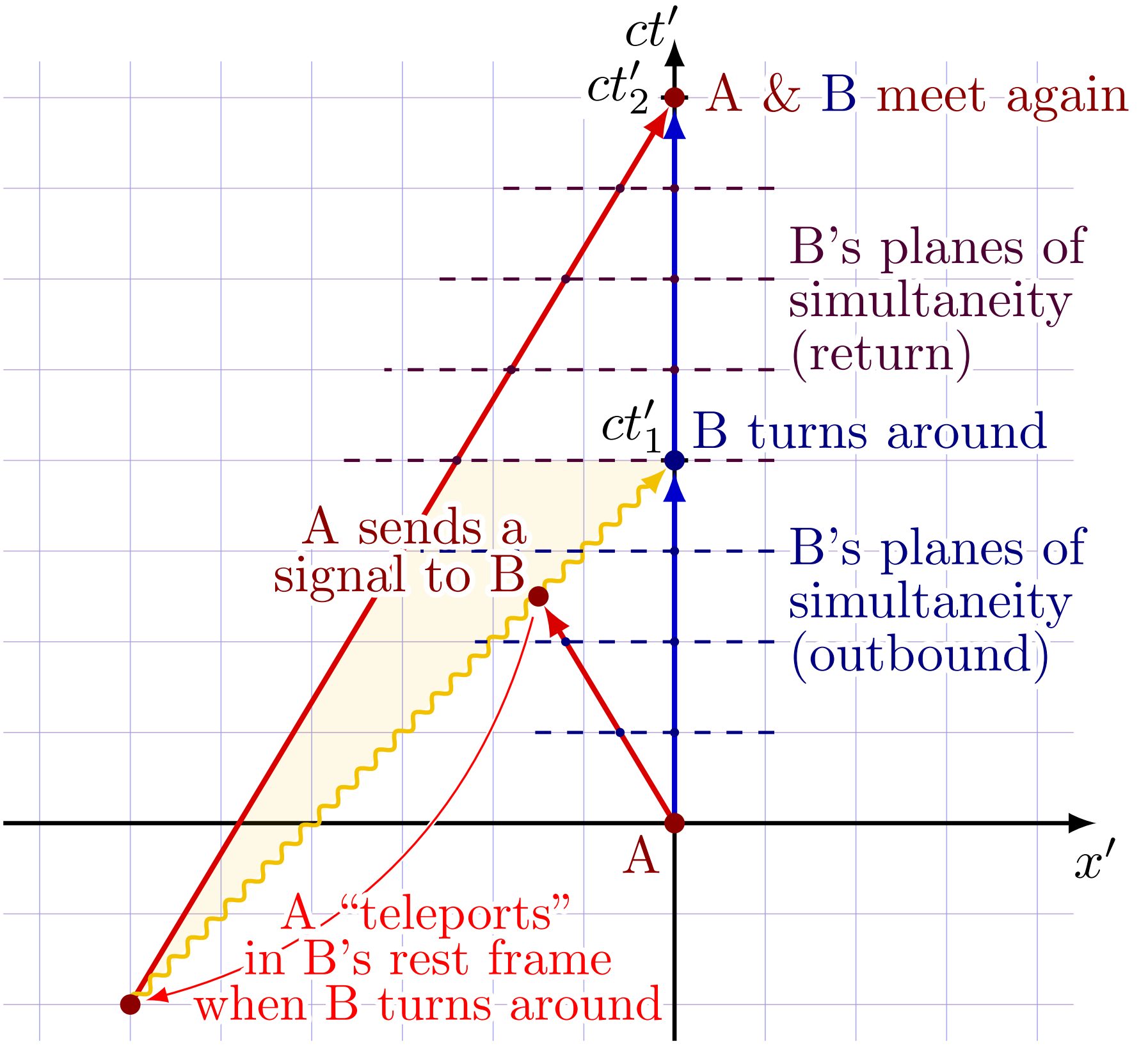

ct
ct'
left door closed in S′
right door closed in S′
both doors closed in S
x'
L_0
L
barn
$L < w < L_0$
x

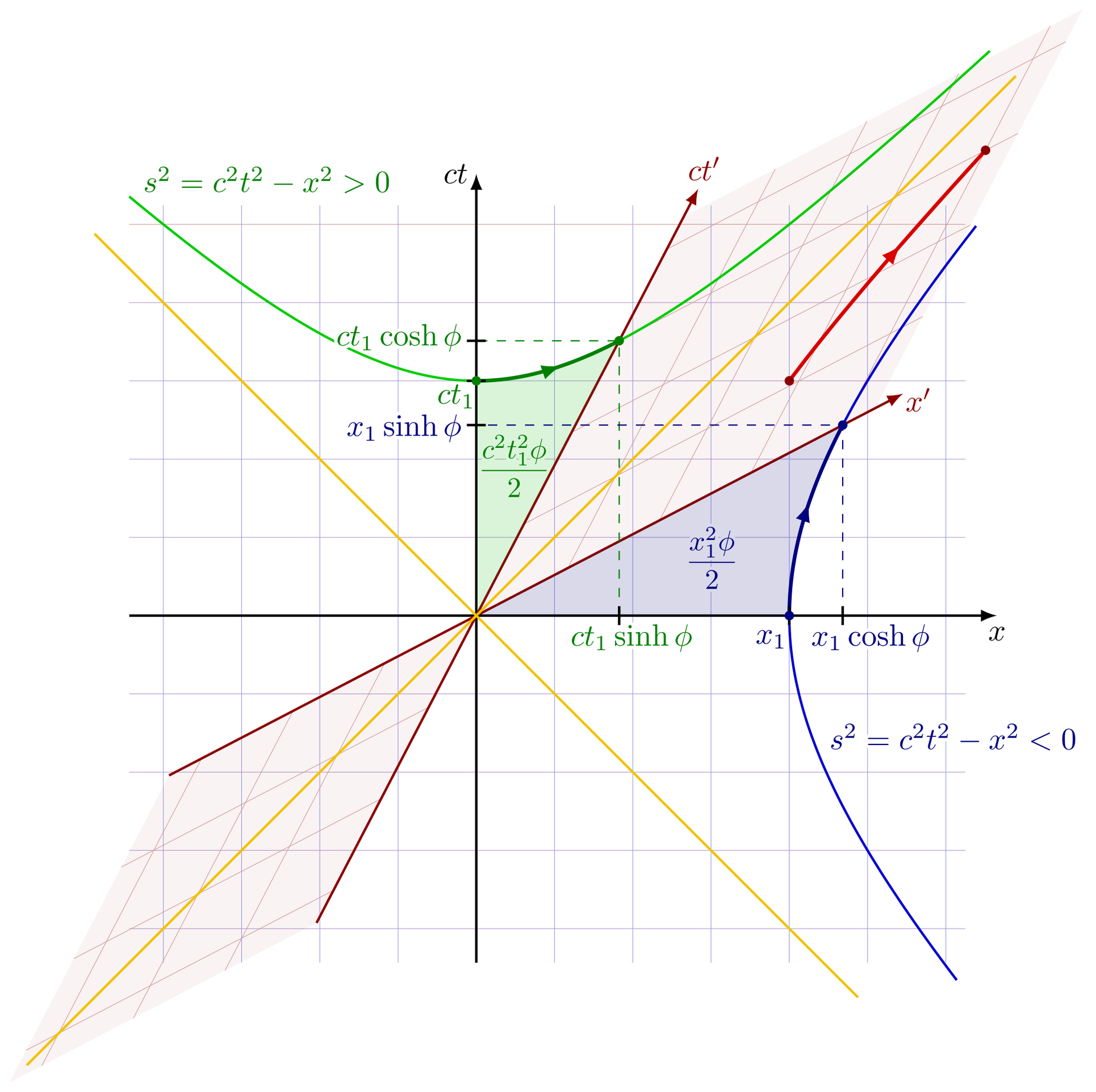

ct
ct′
rod
left door closed in S′
right door closed in S′
both doors closed in S
L_0
barn
x′
L
x

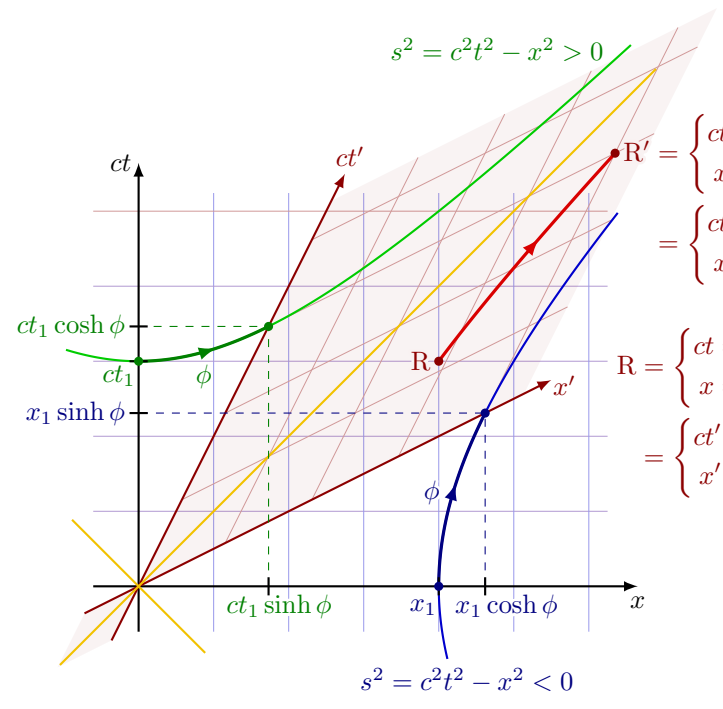

ct
ct_2
A & B meet again
B's planes of simultaneity (return)
$\frac{ct_2}{2} = ct_1$
B turns around
B's planes of simultaneity (outbound)
B leaves
A
x


ct
ct_2
A & B meet again
B's planes of simultaneity (return)
$\frac{ct_2}{2} = ct_1$
B turns around
B's planes of simultaneity (outbound)
A sends a signal to B
B leaves
A
x


ct'
A & B meet again
B's planes of simultaneity (return)
B turns around
A sends a signal to B
B's planes of simultaneity (outbound)
A
x'


ct'
ct'_2
A & B meet again
B's planes of simultaneity (return)
ct'_1
B turns around
A sends a signal to B
B's planes of simultaneity (outbound)
A
x'
A "teleports" in B's rest frame when B turns around


$s^2 = c^2t^2 - x^2 > 0$
ct
ct'
$ct_1 \cosh\phi$
ct_1
$x_1 \sinh\phi$
$\frac{c^2t_1^2\phi}{2}$
x'
$\frac{x_1^2\phi}{2}$
$ct_1 \sinh\phi$
x_1
$x_1 \cosh\phi$
x
$s^2 = c^2t^2 - x^2 < 0$

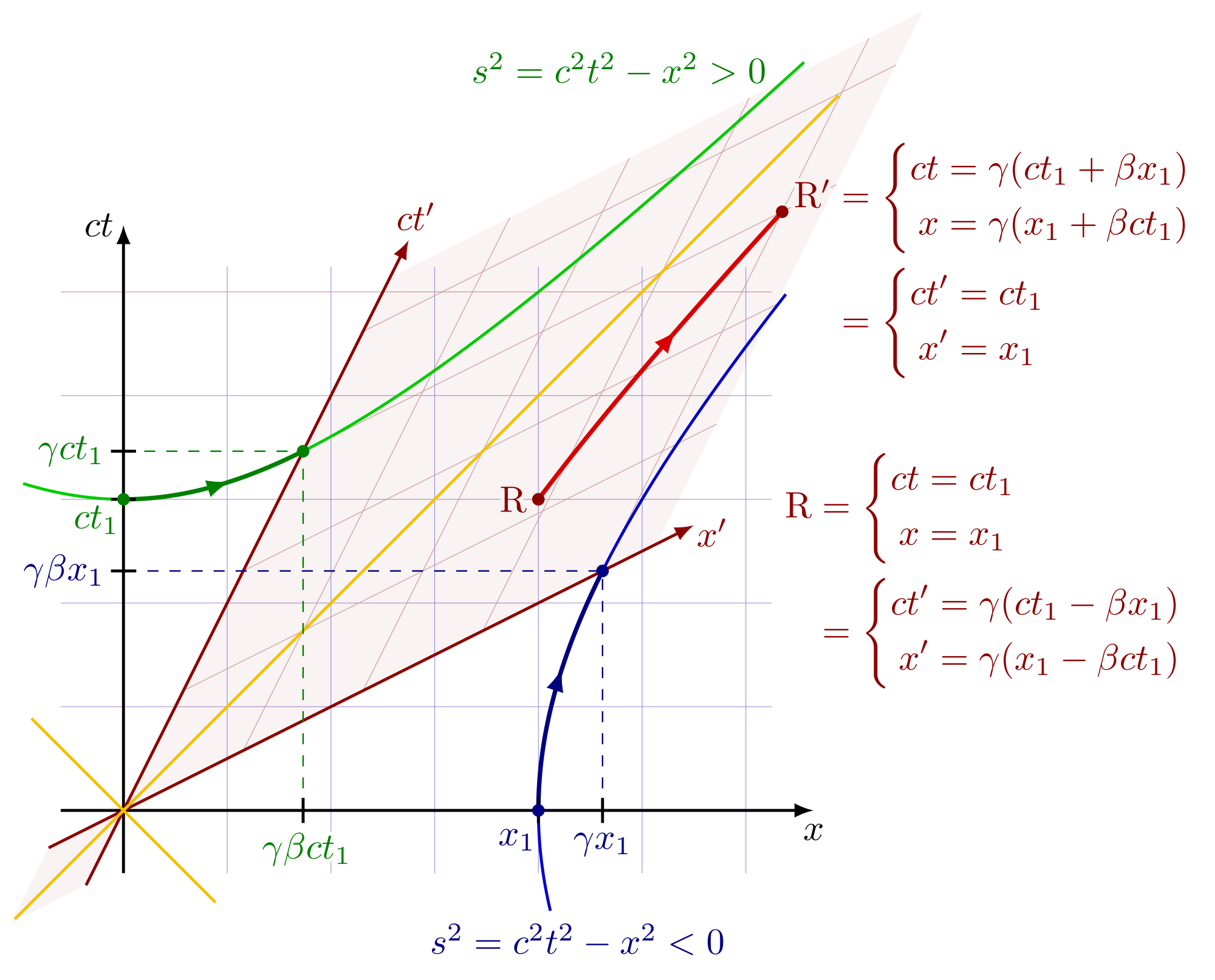


$$\mathrm{R}' = \begin{cases} ct = ct_1 \cosh\phi + x_1 \sinh\phi \\ x = x_1 \cosh\phi + ct_1 \sinh\phi \end{cases}$$

$$= \begin{cases} ct' = ct_1 \\ x' = x_1 \end{cases}$$

$$\mathrm{R} = \begin{cases} ct = ct_1 \\ x = x_1 \end{cases}$$

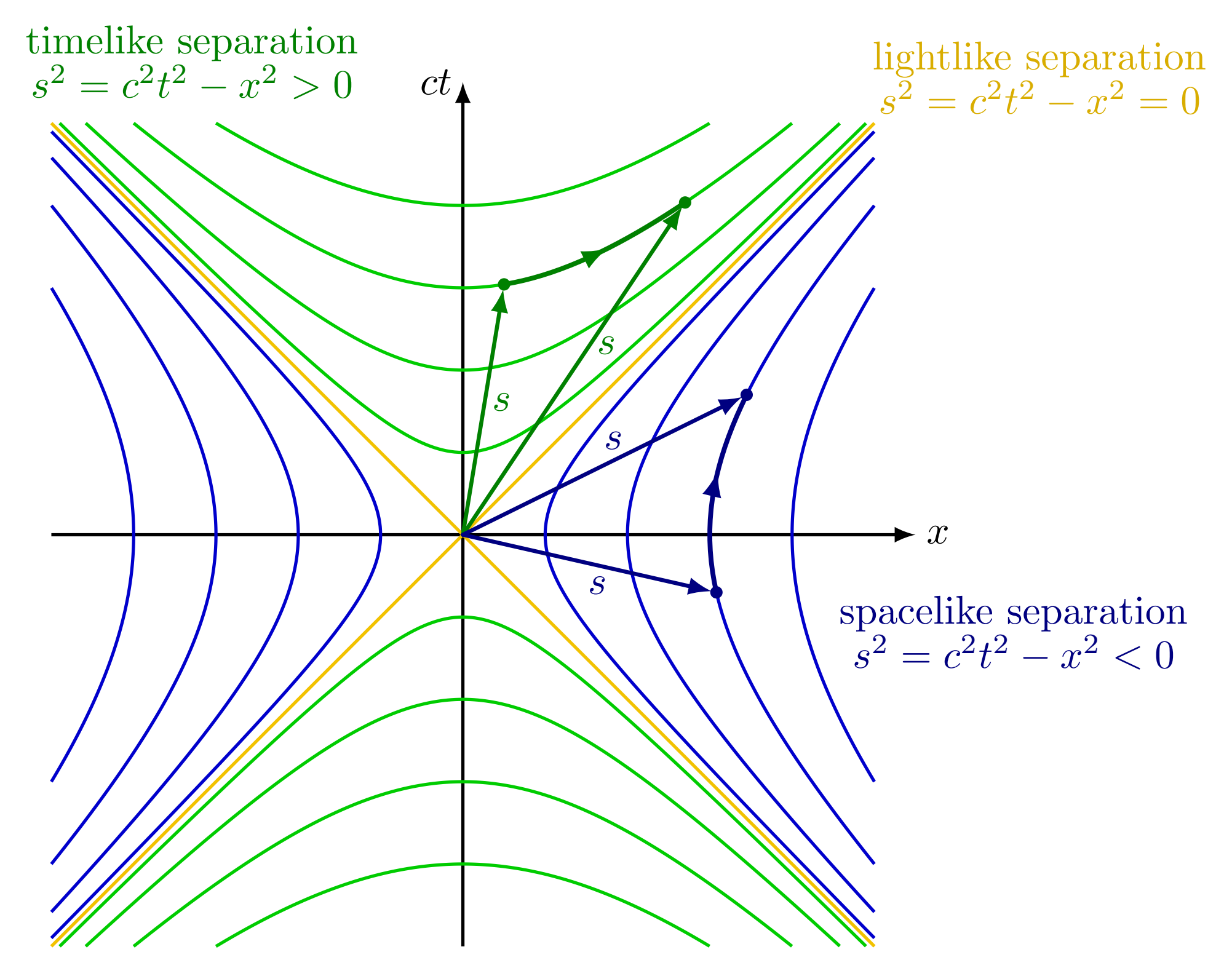
$$= \begin{cases} ct' = ct_1 \cosh\phi - x_1 \sinh\phi \\ x' = x_1 \cosh\phi - ct_1 \sinh\phi \end{cases}$$


$s^2 = c^2t^2 - x^2 > 0$
ct
ct'
γct_1
ct_1
$\gamma\beta x_1$
R
x'
$\gamma\beta ct_1$
x_1
γx_1
x
$s^2 = c^2t^2 - x^2 < 0$
$\mathrm{R}' = \begin{cases} ct = \gamma(ct_1 + \beta x_1) \\ x = \gamma(x_1 + \beta ct_1) \end{cases}$
$= \begin{cases} ct' = ct_1 \\ x' = x_1 \end{cases}$
$\mathrm{R} = \begin{cases} ct = ct_1 \\ x = x_1 \end{cases}$
$= \begin{cases} ct' = \gamma(ct_1 - \beta x_1) \\ x' = \gamma(x_1 - \beta ct_1) \end{cases}$

Lorentz transformation in the x direction

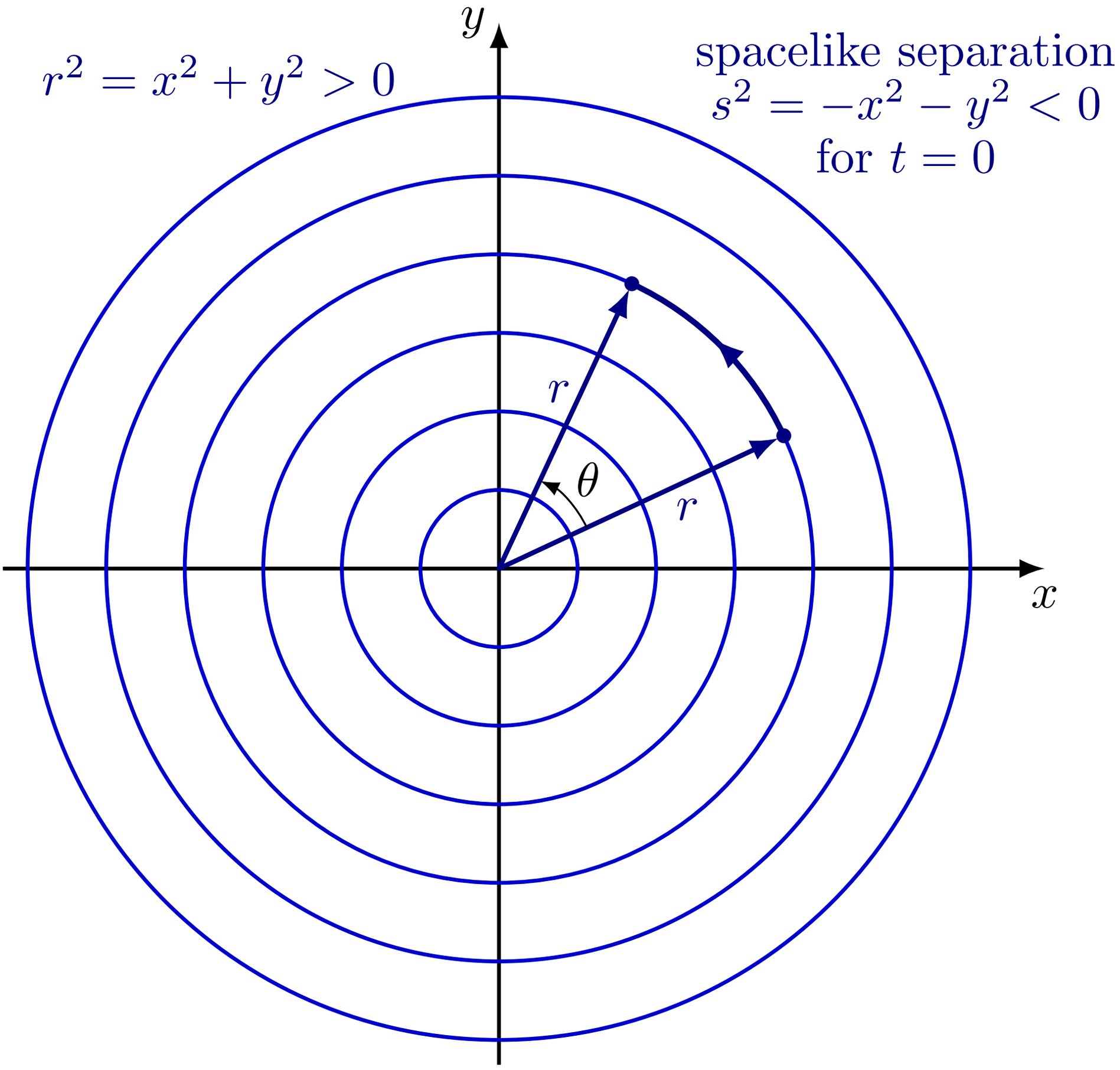
$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ zw \end{pmatrix}$$

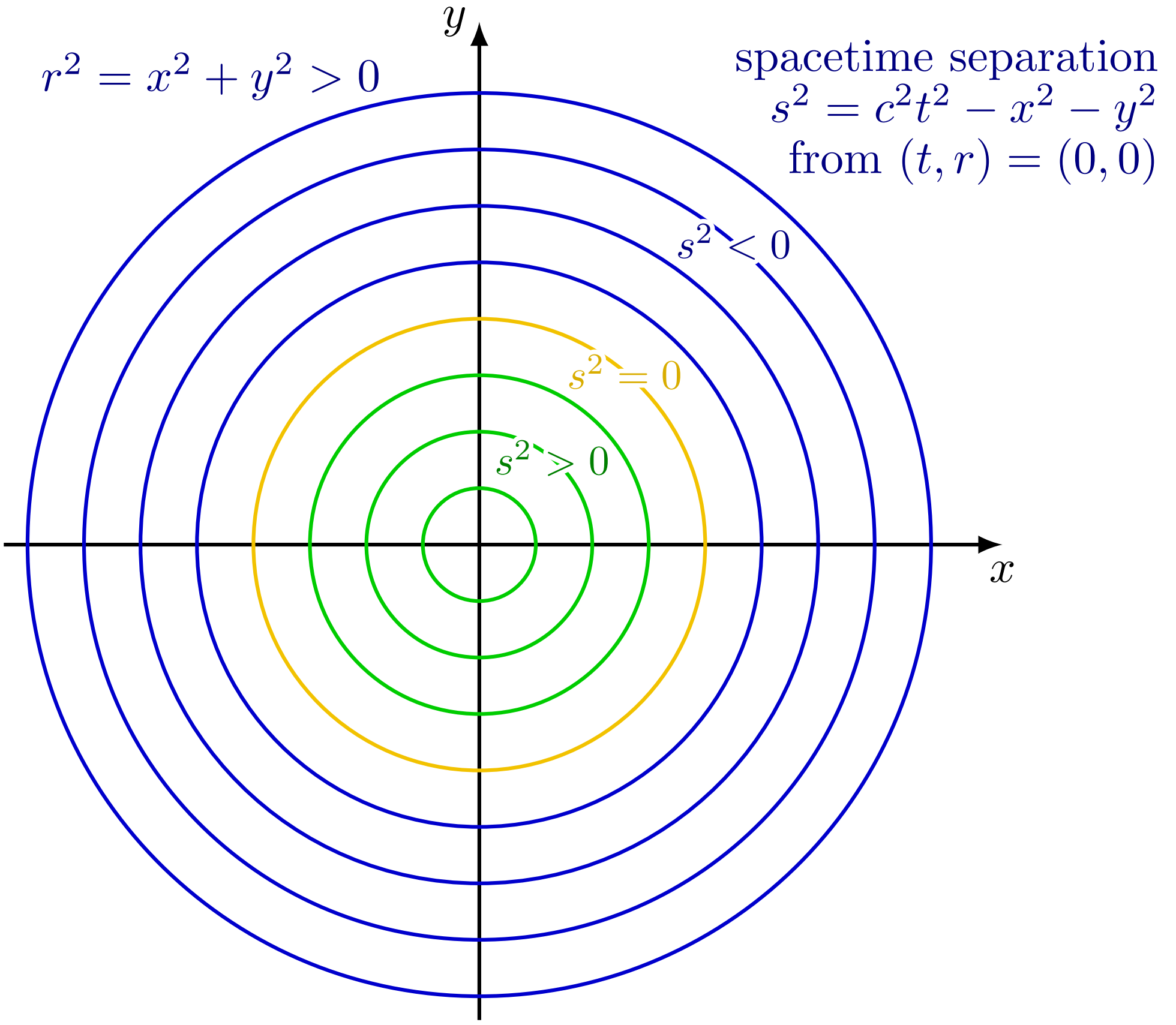
$$= \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

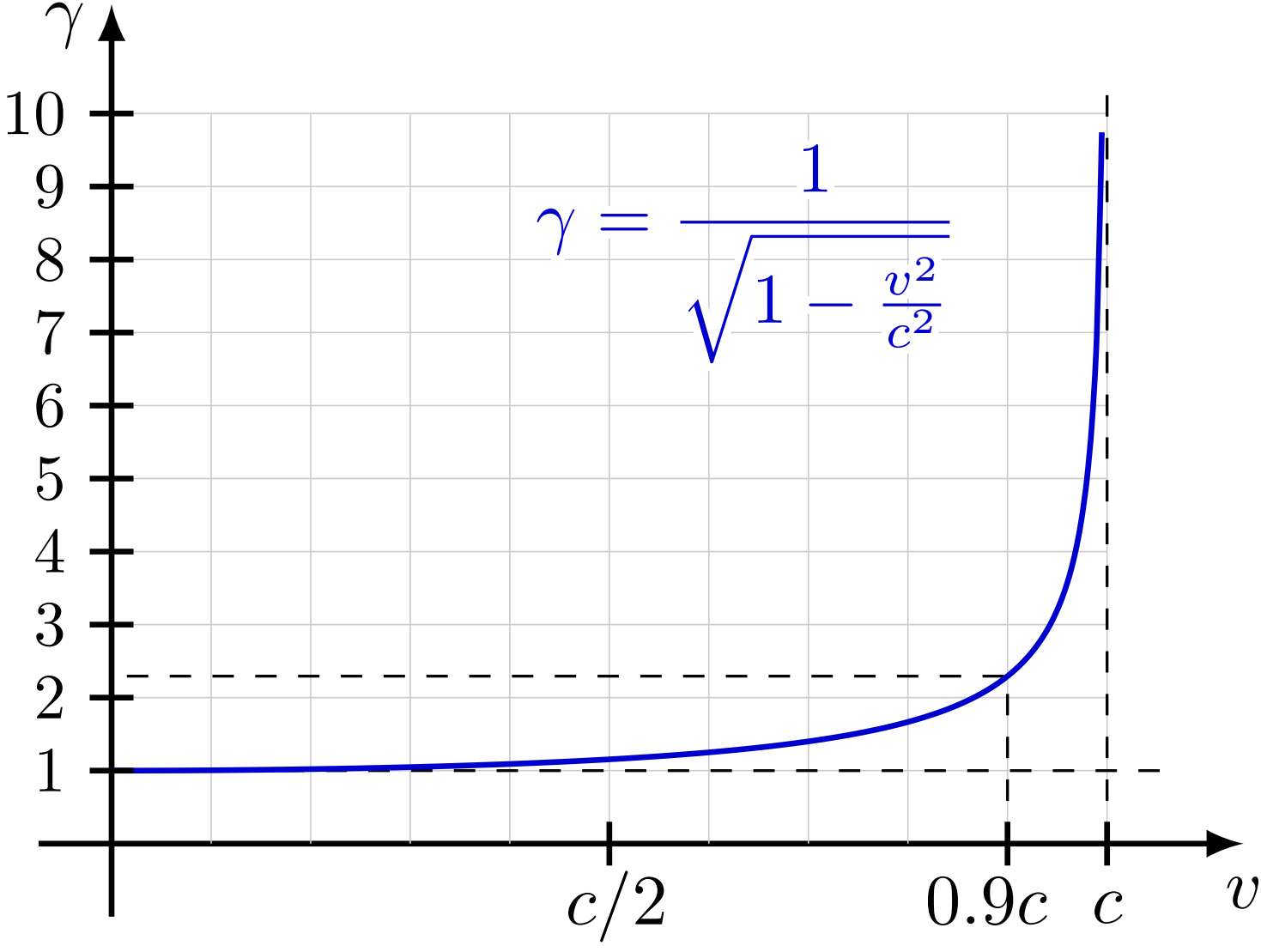

timelike separation
$s^2 = c^2t^2 - x^2 > 0$
lightlike separation
$s^2 = c^2t^2 - x^2 = 0$
ct
x
s
s
s
s
spacelike separation
$s^2 = c^2t^2 - x^2 < 0$

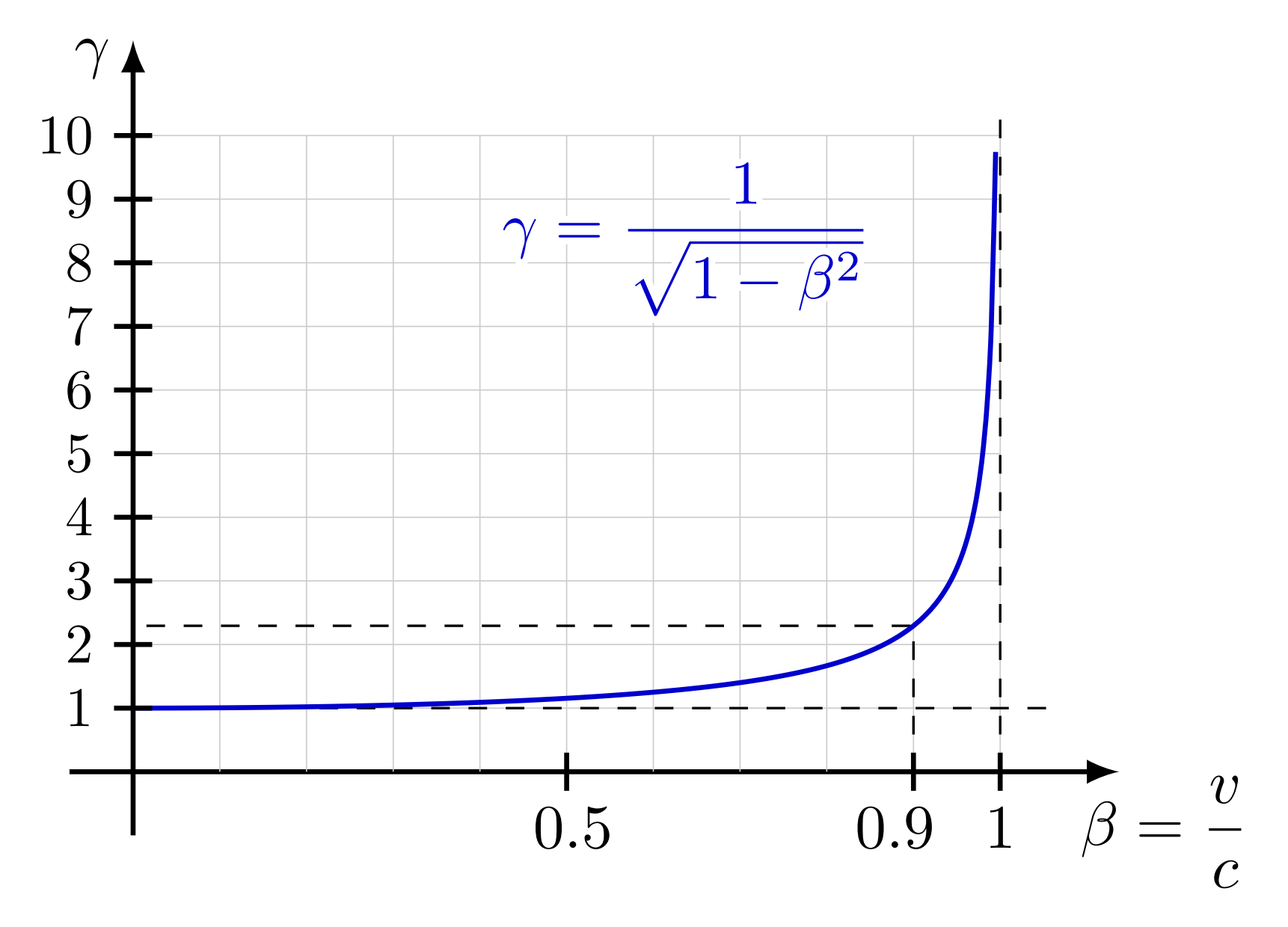
Counterclockwise rotation in the xy plane

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$


$r^2 = x^2 + y^2 > 0$
spacelike separation
$s^2 = -x^2 - y^2 < 0$
for $t = 0$
y
x
r
θ
r


$r^2 = x^2 + y^2 > 0$
spacetime separation
$s^2 = c^2t^2 - x^2 - y^2$
from $(t, r) = (0, 0)$
y
x
$s^2 < 0$
$s^2 = 0$
$s^2 > 0$


γ
10
9
8
7
6
5
4
3
2
1
$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$
c/2
0.9c
c
v


γ
10
9
8
7
6
5
4
3
2
1
$\gamma = \frac{1}{\sqrt{1-\beta^2}}$
0.5
0.9
1
$\beta = \frac{v}{c}$